



Evaluation and prediction of pharmacokinetics of PFOA and PFOS in the monkey and human using a PBPK model

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ARTICLE INFO

Article history:

Received 18 August 2010

Available online 17 December 2010

Keywords:

Physiologically-based pharmacokinetic modeling

PBPK

Toxicokinetic modeling

Perfluoroalkyls

Pharmacokinetics

PFOA

PFOS

ABSTRACT

Perfluoroalkyl acid carboxylates and sulfonates (PFAAs) have many consumer and industrial applications. The persistence and widespread distribution of these compounds in humans have brought them under intense scrutiny. Limited pharmacokinetic data is available in humans; however, human data exists for two communities with drinking water contaminated by PFAAs. Also, there is toxicological and pharmacokinetic data for monkeys, which can be quite useful for cross-species extrapolation to humans. The goal of this research was to develop a physiologically-based pharmacokinetic (PBPK) model for PFOA and PFOS for monkeys and then scale this model to humans in order to describe available human drinking water data. The monkey model simulations were consistent with available PK data for monkeys. The monkey model was then extrapolated to the human and then used to successfully simulate the data collected from residents of two communities exposed to PFOA in drinking water. Human PFOS data is minimal; however, using the half-life estimated from occupational exposure, our model exhibits reasonable agreement with the available human serum PFOS data. It is envisioned that our PBPK model will be useful in supporting human health risk assessments for PFOA and PFOS by aiding in understanding of human pharmacokinetics.

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1. Introduction

Perfluorooctanoate (PFOA) and perfluorooctane sulfonate (PFOS) are fully fluorinated man-made chemicals that have a wide spectrum of industrial and consumer uses. These compounds are used as surfactants in situations requiring stability towards heat and chemical degradation or other unique properties imparted by the perfluorinated chain (Lau et al., 2007). They can also be formed from the degradation (environmental or metabolic) of certain fluorinated materials (Olsen et al., 2007, 2003a). Due to the high electronegativity of the fluorine atom, the C–F bond has a significant dipole moment, imparting ionic character to the bond through partial charges; the partial charges are attractive, making the C–F bond one of the strongest in organic chemistry (Lau et al., 2007). The stability imparted to these compounds by the C–F bond is very desirable industrially; they are stable in air at high temperature, nonflammable, and not readily degraded by strong acids, bases, or oxidizers (Lau et al., 2007). The C–F bond also provides these compounds with low surface tension making them ideal

surfactants (Lau et al., 2004). However, the same properties that make these compounds so useful make them resistant to breakdown by metabolism, hydrolysis, photolysis, or biodegradation and thus they can be persistent in the environment indefinitely. Industrial use of PFOA began in the late 1940s, and since then it has been estimated that cumulative global emissions are between 2400 and 5200 metric tons (Prevedouros et al., 2006). Production and use of PFOS and materials which can form PFOS were initially greater than that of PFOA, but since the phaseout of PFOS by its major manufacturer between 2000 and 2002, global production has dropped (175 metric tons in 2003) (3M, 2003).

Although exposure sources and routes for humans are not well understood, perfluoroalkyl carboxylates and sulfonates (collectively perfluoroalkyl acids or PFAAs) escape to the environment through production, manufacture, disposal, and degradation of other fluoropolymers (Andersen et al., 2008). These compounds are now widespread throughout the global ecosystem. They have been found in wildlife samples from remote ocean sites and the Arctic, although concentrations are usually greater in animals living in more populated and industrial areas (Lau et al., 2007). These compounds have also been identified in parts-per-billion levels in human serum from the general population and in parts-per-million levels for occupationally exposed workers and other populations with high exposure to these compounds (Calafat et al., 2007b; Emmett et al., 2006; Olsen et al., 2007). The long plasma

Abbreviations: PFAAs, collectively refers to perfluoroalkyl acid carboxylates and sulfonates; PBPK model, physiologically based pharmacokinetic model; PFOA, perfluorooctanoate; PFOS, perfluorooctane sulfonate.

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half-lives of PFOA and PFOS (3–5 years) observed in humans is of particular concern because this indicates that they can accumulate, which may result in higher body burdens and potential adverse health effects. Possible exposure sources for humans include drinking water, dust in homes, food and food packaging, fabrics, carpeting, and cookware (Fromme et al., 2009; Ostertag et al., 2009; Sinclair et al., 2007; Skutlarek et al., 2006; Strynar and Lindstrom, 2008; Vestergren and Cousins, 2009; Washburn et al., 2005; Wilhelm et al., 2008).

The pharmacokinetic properties of PFOA and PFOS have been studied. In summary, animal studies have shown that these compounds are well absorbed orally, poorly eliminated, and not metabolized (Johnson et al., 1984; Kuslikis et al., 1992; Ophaug and Singer, 1980; Vanden Heuvel et al., 1991). They are distributed mainly to the serum, liver, and kidney, usually with liver concentrations being 3–5 times higher than serum concentrations in the rat, while liver concentrations in humans and non-human primates are only ~1.3–2 times higher than in serum (DePierre, 2009; Hundley et al., 2006; Johnson et al., 1979; Kemper, 2003; Kudo et al., 2007; Olsen et al., 2003b; Seacat et al., 2002). Distribution at steady state is mainly extracellular (Butenhoff et al., 2004a; Noker and Gorman, 2003). Both compounds have a high affinity for binding to albumin and are thus highly bound in plasma, and to a lesser extent, β -lipoproteins and liver fatty acid binding protein (Han et al., 2003; Kerstner-Wood et al., 2003; Luebker et al., 2002). Both compounds have been found in rodent fetuses as well in as umbilical cord blood of humans and in breast milk indicating that they can cross the placenta and partition into milk, exposing the fetus and neonate (Apelberg et al., 2007; Hinderliter et al., 2005; Karrman et al., 2007; Kuklenyik et al., 2004; Thibodeaux et al., 2003). The most notable aspect of PFOA and PFOS PK is that they exhibit major differences in plasma half-lives across species, and for PFOA, a sex difference in elimination is observed in rats. The elimination half-life for PFOA in female rats is 2–4 h; it is 4–6 days in male rats (Hanhijarvi et al., 1987; Kemper, 2003; Kudo et al., 2002). The gender difference in elimination is age-dependent and the slower elimination in males is observed starting around 3–5 weeks in age (Hinderliter et al., 2006). This gender difference in elimination is not observed for PFOS in rats (half-life is ~30–50 days, depending on the isomer) or mice (half-life is ~40 days) (Benskin et al., 2009; Butenhoff, 2010). Monkeys have half-lives of 21–30 days for PFOA and 4–6 months for PFOS while human half-lives are even longer for both compounds (3–5 years) (Bartell et al., 2010; Butenhoff et al., 2004a; Noker and Gorman, 2003; Olsen et al., 2007; Seacat et al., 2002). Neither the monkey nor human appears to have any major sex differences in elimination of either compound, and this appears to be true for the mouse also (Butenhoff et al., 2004a; Noker and Gorman, 2003; Olsen et al., 2007; Rodriguez et al., 2009; USEPA, 2005). The cause of the differential elimination across sexes and species has not been conclusively demonstrated, but there is evidence from studies performed both in rats and in vitro with rat and human transporters that transporters in the proximal tubule of the kidney may be responsible (Andersen et al., 2006; Kudo et al., 2002; Tan et al., 2008; Weaver et al., 2009; Yang et al., 2010, 2009).

Potential toxicities in animals have been characterized. Subchronic exposure to PFOS led to significant weight loss, reductions in serum cholesterol and thyroid hormones, and hepatotoxicity. Developmental toxicity studies with PFOS in rats, mice, and rabbits reveal reduction of fetal weight, cleft palate, and reduced neonatal survival (Lau et al., 2004). Repeated-dose studies with PFOA in rodents showed induction of liver tumors, Leydig cell tumors, and pancreatic acinar cell tumors (Andersen et al., 2008; Biegel et al., 2001; USEPA, 2005). Reproductive toxicity studies with PFOA in rodents showed increased post weaning mortality, decreased body weight, and delayed sexual maturation (Butenhoff et al., 2004b).

Both compounds are agonists for peroxisome proliferator activated receptor alpha (PPAR- α) and the tumor induction has been attributed to this activation (Elcombe et al., 2010). However, PPAR- α activation may not be of human relevance because humans express much lower levels of these receptors than do rodents and human liver cells have been shown to be much less responsive to the effects of ammonium PFOA than rat liver cells (Bjork and Wallace, 2009; Lake, 2009).

Due to their prevalence and stability in the environment, toxic effects observed in animal studies, and long half-life in humans, PFAAs have drawn considerable attention from public and regulatory agencies with regard to health risks that they may present. Risk assessments and interpretation of available human data are slowed due to lack of framework to understand and estimate human pharmacokinetics. Previously, a biologically based compartmental model was developed by our group that was able to describe the complex PK of PFOA and PFOS in rats and monkeys (Andersen et al., 2006; Tan et al., 2008). The key process required to describe kinetics was resorption by renal transporters in the filtrate compartment. The goal of the present work was to develop a complete PBPK model for the monkey for PFOA and PFOS and then scale up to the human. The consistency of renal resorption to describe the kinetics in both the rat and monkey suggests the existence of a saturable, high-affinity resorption process that governs the kinetics of these compounds in other species, including humans (Tan et al., 2008). If confirmed, once the transporter protein that is responsible for renal resorption of PFOA and PFOS is identified, information on the activity of that transporter can then be used to identify susceptible sub-populations due to polymorphic variation. The human model was used to successfully simulate available data and also to predict and thus examine variability in human half-lives of PFOA. The result of these efforts is a human PBPK model with reasonable predictive ability that can be used to aid in risk assessment for perfluorinated compounds.

2. Methods

2.1. Monkey model development

The structure for the monkey model was developed from a biologically based compartmental model for PFOA and PFOS in rats and monkeys (Fig. 1) (Andersen et al., 2006; Tan et al., 2008). The compartmental model contained compartments for plasma, liver, tissues, and filtrate, and it was developed to examine the role of renal resorption through a saturable transport process that is thought to be responsible for the long half-life of PFOA and PFOS in rats and monkeys. Free chemical in the central (plasma) compartment can move to the liver compartment, tissue compartment, or be filtered by the filtrate (i.e., kidney) compartment. Once in the filtrate compartment, the chemical can be resorbed back to the plasma compartment by a saturable process with transporter maximum T_m and affinity constant K_t . This model was able to successfully describe PFOA and PFOS kinetics from oral and IV dosing in rats and monkeys, but in order to extrapolate the model to the human to aid in risk assessment, a physiologically based model with realistic tissue volumes and blood flows and physicochemical and biochemical properties of the chemical under study were needed. Thus, a physiologically-based pharmacokinetic (PBPK) model was developed for PFOA and PFOS in the monkey for use in extrapolation to the human.

2.1.1. Study designs and data – PFOA in monkeys

Pharmacokinetic data for PFOA in the monkey have been reported by Butenhoff et al. (2004a) for oral and IV dose routes. Both data sets were used to develop the current model. In the IV dosing

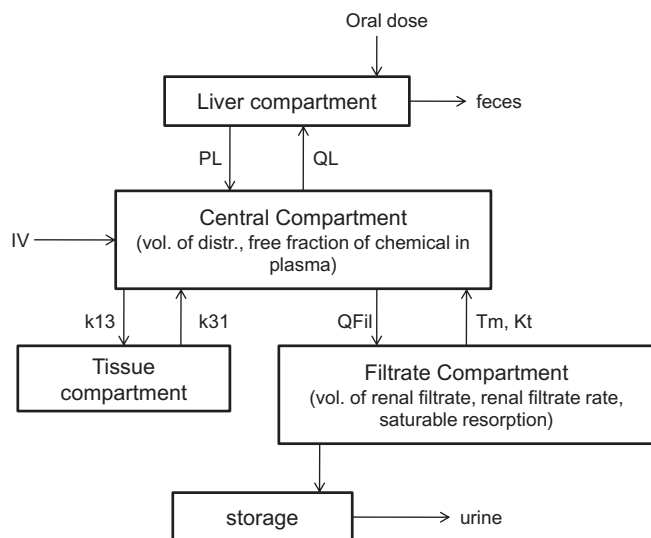


Fig. 1. Structure of biologically motivated model for PFOA and PFOS in rats and monkeys. Figure adapted from Tan et al. (2008). Free chemical in the central compartment can move to the liver compartment based on flow rate QL and partition coefficient PL, to the tissue compartment based on the rate constants k13 and k31, or be filtered by the filtrate compartment with flow rate Qfil. Chemical can be resorbed back to the plasma compartment with transporter maximum Tm and affinity constant Kt.

study, male and female cynomolgus monkeys (3/sex) received a single IV dose of 10 mg/kg PFOA and were then monitored for 123 days after dosing. PFOA concentration in plasma was measured at 0.5, 2, 4, 8, and 24 h post dose and on days 2, 4, 7, 11, 14, 21, 28, 57, 79, 87, and 123. No adverse effects related to PFOA were reported (Butenhoff et al., 2004a). In the oral dosing study, male cynomolgus monkeys were administered a daily oral dose by capsule of 0 ($n = 6$), 3 ($n = 4$), 10 ($n = 6$), or 20 mg/kg ($n = 6$) PFOA for 6 months. Several of the animals were then allowed to recover and were monitored for 90 or more days after dosing cessation ($n = 2$ for the 10 mg/kg dose group and $n = 2$ in the 20 mg/kg dose group). Monkeys in the 20 mg/kg dose group were originally started on 30 mg/kg PFOA; however, dosing had to be stopped after 12 days due to toxicity. Dosing began again at day 21, and the dose was lowered to 20 mg/kg. Dosing again had to be discontinued for three of the monkeys due to toxicity on day 43, 66, and 81. Only two monkeys in the high-dose group received treatment for the entire 6 months. During the dosing period, PFOA concentration in plasma and urine of the monkeys still receiving doses were taken at weeks 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, and 26. During recovery, plasma and urine PFOA concentrations were measured for only the 10 and 20 mg/kg dose groups; these were measured once every two weeks from weeks 28 through 40 of the study. Because there was no recovery data for the 3 mg/kg dose group, and the monkeys in the 30/20 mg/kg dose group exhibited signs of toxicity and had an irregular dosing regimen, only the plasma and urine data from the 10 mg/kg dose group was used for development of the monkey model for PFOA by the oral route. Also, although the 10 mg/kg dose would be a much higher concentration than humans would be exposed to environmentally, the lower dose and the effects and PK associated with it would be more relevant for development of a human model. The monkey model was used to simulate the data from both the IV and oral studies with PFOA in the cynomolgus monkey.

2.1.2. Study designs and data – PFOS in monkeys

Similar to PFOA, there is existing PK data for PFOS in cynomolgus monkeys for IV and oral dose routes. Both data sets were used

to develop the current monkey model. The IV data has been reported by Noker and Gorman (2003). Male and female monkeys (3/sex) were administered a single IV dose of 2 mg/kg PFOS and then monitored for 161 days after dosing. PFOS concentrations in plasma and urine of the monkeys were measured 0.5, 2, 4, 8, and 24 h post dose and on days 2, 7, 14, 21, 28, 42, 56, 70, 105, and 161 post dose. The other data used for model development was from a subchronic toxicity study of PFOS reported by Seacat et al. (2002). Male and female monkeys received daily oral doses (by capsule) of 0.03, 0.15, or 0.75 mg/kg PFOS for 26 weeks. Two monkeys per sex in the 0.15 and 0.75 mg/kg dose groups were allowed to recover and monitored for a year. Plasma concentrations of PFOS were measured throughout the dosing period for all 3 dose groups and during the recovery period for the mid- and high-dose groups. Significant adverse effects were observed only in the high-dose group, which included death of 2 of 6 male monkeys for which PFOS exposure could not be ruled out as a contributing cause. The monkey model was used to simulate the data from both the IV and oral studies with PFOS in the cynomolgus monkey.

2.2. Model structure – monkeys

The PBPK model structure for PFOA and PFOS in the monkey is the same (Fig. 2). The model contains compartments for plasma, gut, liver, kidney, filtrate, fat, skin, and a lumped compartment for remaining body tissues. The plasma and liver are the primary target tissues for PFOA and PFOS with possible involvement of enterohepatic circulation (Butenhoff et al., 2004a; Johnson et al., 1979; Kemper, 2003; USEPA, 2005). These compounds also activate nuclear receptors PPAR- α and CAR/PXR known to be associated with a hypertrophic response in liver tissue (Elcombe et al., 2010; Rosen et al., 2010). The liver in some cases is the site of accumulation (Andersen et al., 2008; Kennedy et al., 2004; Lau et al.,

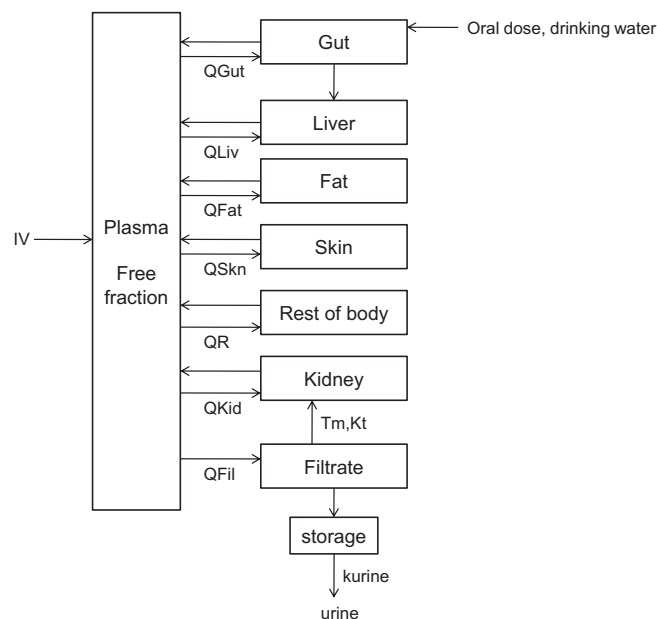


Fig. 2. Structure of PBPK model for PFOA and PFOS in monkeys and humans. Chemical is taken up into the plasma (iv) or into the gut (oral). From the gut, chemical is transported to the liver by the portal blood. Only the free fraction of chemical in plasma is assumed to be available for partitioning into tissues. Chemical is eliminated through the filtrate compartment to storage into urine. While in the filtrate compartment, chemical can be reabsorbed back into the plasma through a saturable process with a transporter maximum Tm and affinity constant Kt. The Qs indicate blood flows into and out of tissues. Qfil is not a blood flow; it is a clearance (L/h) from the plasma to the filtrate compartment.

2004; Luebker et al., 2002; USEPA, 2005). The PBPK model retains the same features that were important for describing kinetics of PFOA and PFOS in the biologically based model of Andersen and Tan, namely, a free fraction of chemical in the plasma and renal resorption in the filtrate compartment (Andersen et al., 2006; Tan et al., 2008). The fat and skin compartments were built into the monkey model because these compartments will be of interest in the human. PFOA and PFOS are lipophobic, thus fat would serve as an exclusionary volume for these chemicals, and dermal absorption is a possible exposure route for the human (USEPA, 2005). An IV dose to the monkey enters the plasma, and oral doses enter the gut and are then passed to the liver by the portal blood. Only the free fraction of chemical in plasma is assumed to be available to partition into tissues. Chemical is eliminated through the filtrate compartment to storage and subsequently to urine. The clearance from plasma to the filtrate compartment is described by a flow rate, Q_{fil} . In the filtrate compartment, chemical can be reabsorbed by a saturable process with a transporter maximum (T_m) and affinity constant (K_t) into the kidney and back into systemic circulation. Coding and simulations for both the PFOA and PFOS monkey models was performed in the Berkeley–Madonna program (Macey et al., 2000). Model code can be found in the Supplementary Material.

2.3. Model parameterization

2.3.1. Chemical-specific parameters for PFOA and PFOS

Chemical-specific parameters used for the monkey PFOA and PFOS models are presented in Table 1. Tissue:plasma partition coefficients for PFOA for all tissues in the model were estimated from tissue concentration data reported by Kudo et al. (2007) for a single IV dose of PFOA to male Wistar rats. Tissue:blood partition coefficients for PFOS for all tissues in the model were estimated

from tissue concentration data from male C57Bl/6 mice (mice were fed PFOS in their diet from one to five days) (DePierre, 2009). PFOA and PFOS have been shown to be >97% bound to plasma proteins such as albumin in the rat, monkey, and human (Han et al., 2003; Jones et al., 2003; Kerstner-Wood et al., 2003); a constant free fraction of each chemical (<0.03) was used. Because of the limited evidence of extensive binding for PFOA and PFOS, it was necessary to estimate the free fraction of chemical in plasma (Free) for each compound to provide the best fit the plasma concentration data from the IV and oral studies while adhering to the aforementioned constraint.

It is hypothesized that the long plasma half-lives of PFOA and PFOS are due to renal resorption by transporters in the proximal tubule of the kidney (Andersen et al., 2006; Tan et al., 2008). In fact, more recent research has suggested that renal tubular resorption transporters are capable of transporting PFOA (Nakagawa et al., 2009; Weaver et al., 2009; Yang et al., 2010, 2009). In order to describe the kinetics of these chemicals, a description of saturable renal resorption was implemented in the filtrate compartment in the model. The parameters used to describe renal resorption are the transporter maximum (T_m) and the transporter affinity constant (K_t). In the rat, a significant sex difference is observed in the clearance of PFOA but not PFOS (Andersen et al., 2008; Kemper, 2003; Kennedy et al., 2004; Kudo et al., 2002). It has been shown that male rats exhibit higher expression of a particular organic anion transporter, *Oatp1a1*, that is thought to play a role in renal resorption, thus extending the half-life of PFOA in the male rat (Weaver et al., 2009; Yang et al., 2009). However, no significant sex differences are observed in elimination of either PFOA or PFOS in the monkey (Butenhoff et al., 2002, 2004a; Seacat et al., 2002), both chemicals have long half-lives in both sexes of the monkey (20–30 days for PFOA; 5–6 months for PFOS). Therefore, in the monkey model, the male and female monkeys have the same

Table 1
Chemical parameters for PBPK models for PFOA and PFOS in monkeys and humans.

PFOA	Definition and units	Value	Source
T_m	Resorption maximum (mg/h/kg ^{0.75}) ^a	0.15 (monkey); 10 (human; 3.8 yrs); 6 (human (2.3 yrs)	Fit to plasma concn. in monkey; estimated to give desired half-life in human (3.8 or 2.3 yrs)
K_t	Affinity constant (mg/L)	0.055	Fit plasma and urine concn. in monkey
Free	Free fraction of chemical in plasma	0.02	Fit to plasma concn. in monkey
PL	Liver:plasma partition coefficient	2.2	Rat tissue data (Kudo et al., 2007)
PF	Fat:plasma partition coefficient	0.04	Rat tissue data (Kudo et al., 2007)
PK	Kidney:plasma partition coefficient	1.05	Rat tissue data (Kudo et al., 2007)
PSk	Skin:plasma partition coefficient	0.1	Rat tissue data (Kudo et al., 2007)
PR	Rest of body:plasma partition coefficient	0.12	Rat tissue data (Kudo et al., 2007)
PG	Gut:plasma partition coefficient	0.05	Rat tissue data (Kudo et al., 2007)
Kurine	Urinary elimination rate (/h/kg ^{-0.25})	50 (monkey); $3e^{-4}$ (human)	Fit to urine concn. in monkey and human
PFOS			
T_m	Resorption maximum (mg/h/kg ^{0.75}) ^a	1.3 (monkey); 3.5 (human)	Fit to plasma concn. data in monkey; estimated to give desired half-life in human (5.4 yrs)
K_t	Affinity constant (mg/L)	0.023	Fit to plasma concn. data in monkey
Free	Free fraction of chemical in plasma	0.025	Fit to plasma concn. in monkey
PL	Liver:plasma partition coefficient	3.72	Mouse tissue data (DePierre)
PF	Fat:plasma partition coefficient	0.14	Mouse tissue data (DePierre)
PK	Kidney:plasma partition coefficient	0.8	Mouse tissue data (DePierre)
PSk	Skin:plasma partition coefficient	0.29	Mouse tissue data (DePierre)
PR	Rest of body:plasma partition coefficient	0.2	Mouse tissue data (DePierre)
PG	Gut:plasma partition coefficient	0.57	Mouse tissue data (DePierre)
Kurine	Urinary elimination rate (/h/kg ^{-0.25})	0.003 (monkey); 0.001 (human)	Fit to urine concn. in monkey and human

values for T_m and K_t ; i.e., we assume that the transporter(s) responsible for resorption of PFOA would be the same in both sexes. The same goes for PFOS; however, we assume that either the transporter(s) responsible for resorption of PFOS is different from that of PFOA or simply that the activity of each compound with the transporter(s) is different. PFOS has a longer half-life; the capacity and affinity of the transporter are thus higher than those for PFOA. The transporter maxima and affinity constants for each chemical were estimated by fitting them to the plasma and urine concentration data for PFOA and PFOS in both sexes. The transporter maximum was scaled as $BW^{0.75}$. The urinary elimination rate constant (K_{urine}) was estimated to fit the urine concentration data from the IV doses of chemicals.

2.3.2. Physiological parameters for the monkey model – PFOA and PFOS

The physiological parameters for the monkey can be found in Table 2. Body weights for the monkeys were obtained from Butenhoff et al. (2004a) and Seacat et al. (2002). Fractional tissue volumes were the values reported from Brown et al. (1997) and Forsyth et al. (1968) for rhesus monkeys. Tissue volumes were scaled linearly to body weight. Cardiac output and fractional blood flows to tissues were obtained from Fisher et al. (2000) and Forsyth et al. (1968) for rhesus monkeys. Blood flows were scaled as $BW^{0.75}$. Because PFOA and PFOS do not partition into red blood cells, (Ehresman et al., 2007) blood flows were adjusted for plasma flows to tissues by multiplying the blood flow by 1-hematocrit.

2.4. Model evaluation

For PFOA, we used the monkey model to simulate plasma and urine concentrations of the chemical resulting from IV and oral dose routes. The simulated concentrations were compared to the experimental data of Butenhoff et al. (2002, 2004a). For PFOS, the monkey model was used to simulate plasma concentrations of PFOS from a single IV dose and daily oral doses; the simulated concentrations were then compared to the experimental data of Noker and Gorman (2003) and Seacat et al. (2002).

The adequacy of the monkey models and the corresponding parameters were assessed based on whether the model was able to consistently describe the available pharmacokinetic data on PFOA and PFOS in cynomolgus monkeys. To evaluate the current monkey models, the correspondence between the data points for tissue concentrations of the chemicals and the model-predicted values at different time points were examined visually to make

sure that the simulations were consistent with both the plasma and urine concentrations. The shapes of the simulated curves were assessed to make sure that they were in line with the trends observed in the experimental data. Agreement between the experimental data and the model simulations was good overall. Further efforts to refine the model parameters will require more detailed studies (i.e., plasma protein binding and transporter elucidation in the monkey).

A normalized local sensitivity analysis was performed on the model to examine the influence of each model parameter on the model output. Sensitivity coefficients were calculated for the predicted plasma area under the curve (AUC; total concentration) with the original parameters and for that resulting from a 1% change in each parameter value. The normalized sensitivity coefficients were calculated by the following equation:

$$\text{Sensitivity coefficient} = \frac{\frac{A-B}{B}}{\frac{C-D}{D}}$$

A is the AUC resulting from the 1% increase in parameter value, B is the AUC resulting from the original parameter value, C is the parameter value increased by 1%, and D is the original parameter value. Plasma AUC predictions for the monkeys were run at 3 and 10 mg/kg doses for PFOA and 0.03 and 0.75 mg/kg doses for PFOS.

2.5. Human model development

2.5.1. Study designs and data – human

PFOA and PFOS health and pharmacokinetic data in humans is available from occupational and environmental and epidemiological investigations and worker medical surveillance. Developmental and immunotoxic effects have been reported in studies with rats and mice exposed to PFOA and PFOS. However, fetal effects in the offspring of treated animals were observed only at high doses that caused maternal toxicity (Lau et al., 2004). Similarly, the interpretation of decreased spleen and thymus weights and altered natural killer cell activity in the animals in the immunotoxicity studies were complicated by the high doses used in the studies and also reduced food intake (DeWitt et al., 2009; Dong et al., 2009; Qazi et al., 2010; Yang et al., 2000). Repeat-dose studies in rodents and monkeys with PFOS have shown reduced body weight, increased liver weight, reduced cholesterol, and a steep dose-response curve for mortality (Goldenthal, 1978; Seacat et al., 2002). A chronic bioassay with PFOS in SD rats showed an increase in hepatocellular adenomas at only the highest dose of 20 ppm in the diet (Lau et al., 2007). Two separate chronic bioassays have

Table 2
Physiological parameters for monkey and human PBPK models.

Physiological parameters	Monkey	Human	Source
Cardiac output (L/h/kg ^{0.75}), QCC	15	12.5	Forsyth et al. (1968), Brown et al. (1997)
Blood flow, fraction of CO			
Fat, QFC	0.02	0.052	Fisher et al. (2000), Brown et al. (1997)
Liver, QLC	0.2	0.25	Fisher et al. (2000), Brown et al. (1997)
Kidney, QKC	0.123	0.175	Forsyth et al. (1968), Brown et al. (1997)
Filtrate, QF _{ILC}	0.0246	0.035	20% of kidney blood flow; GFR
Gut, QGC	0.096	0.181	Forsyth et al. (1968), Brown et al. (1997)
Skin, QSkC	0.048	0.058	Forsyth et al. (1968), Brown et al. (1997)
Hematocrit, Htc	0.41	0.44	Davies and Morris (1993), Brown et al. (1997)
BW (kg, average)	3.5–5.5	70	Butenhoff et al. (2004a,b), Seacat et al. (2002), Brown et al. (1997)
Tissue volume, fraction of BW			
Liver, VLC	0.026	0.026	Brown et al. (1997)
Fat, VFC	0.05	0.214	Fisher et al. (2000), Brown et al. (1997)
Kidney, VKC	0.004	0.004	Brown et al. (1997)
Filtrate, VF _{ILC}	0.0004	0.0004	10% of kidney volume
Gut, VGC	0.017	0.017	Brown et al. (1997)
Plasma, VPlasC	0.0448	0.0428	Davies and Morris (1993), Brown et al. (1997)

examined the tumorigenicity of PFOA. In the study by Biegel et al., PFOA was found to increase the incidence of benign hepatocellular, Leydig cell, and pancreatic acinar cell tumors at only a high dose of 300 ppm in the diet (Biegel et al., 2001). In a Riker Pharmaceuticals study, PFOA increased Leydig cell hyperplasia only at a high dietary dose of 300 ppm (Riker, 1987).

However, the PPAR- α mode of action that is responsible for tumor development in rodents is most likely not relevant for humans because PPAR- α activation in humans does not lead to a hyperplastic response (Gonzalez and Shah, 2008; Kennedy et al., 2004; Lake, 2009). PPAR- α activators are non-genotoxic carcinogens, and they include hypolipidemic drugs and cholesterol-lowering drugs. In contrast to the liver tumors observed when these agents are chronically administered to rats and mice, there is no evidence that administration over an extended period of time in humans are associated with liver or any other types of tumors in humans, which indicates a species difference in tumorigenic response (Ashby et al., 1994; Gonzalez and Shah, 2008). Epidemiology studies of exposure to PFOA and PFOS and adverse health events are inconclusive; in fact, no causal associations of adverse health effects to PFOA or PFOS exposure have been observed (Lau et al., 2007; USEPA, 2009). Most of the human data available for PFOA is for a population in Little Hocking, Ohio. The residents of this community get their drinking water from the Ohio River, which is adjacent to a plant in Washington, West Virginia that has manufactured fluoropolymers since 1951. The highest levels of PFOA in public water supplies in the US have been found in the Little Hocking water system (Emmett et al., 2006). Emmett et al. measured the serum PFOA levels in residents of the Little Hocking population as well as the concentration of PFOA in the public drinking water (mean concentration = 3.55 ppb), which is the primary route of exposure for these residents (Emmett et al., 2006). The available data from this study for Little Hocking was from 291 individuals whose primary water source was the public water system. Only a mean, median, and interquartile range is available for the population's serum PFOA concentration; we were not able to obtain individual data. Also, it is unknown for how long the residents were exposed to the contaminated water. No data was collected on serum PFOS concentrations in this study. Aside from Emmett's data, more data for the Little Hocking population is available on the Little Hocking Water Association (LHWA) website (LHWA, 2005). This data contains serum PFOA, PFOS, and several other perfluorinated chemical concentrations for some residents ($N = 25$) as of May 2005 in addition to gender, age, and exposure duration. Both data sets were used to evaluate the predictive ability of the human model.

Another data set from the Little Hocking population used to test the model was from the C8 Health Project, which was conducted as part of a settlement of a class-action lawsuit (Leach v. E.I. du Pont de Nemours & Co., No. 01C-608 (W. Va., Wood County Cir. Ct. filed Apr. 10, 2002)) from August 2005–July 2006. The participants in the study were from Little Hocking and surrounding communities that were served with public drinking water, and the purpose of the study was to assess the serum PFOA levels in the residents. The results of the C8 Health Project were published by Steenland et al. (2009), and the average serum PFOA concentration was ~227 ppb, which is approximately half of what was measured by Emmett et al. (2006) (mean ~448 ppb). The Steenland et al. study did not give a drinking water concentration of PFOA; thus, the range from the Emmett study (1.5–7.2 ppb) was tested for simulating the serum PFOA concentrations. The concentration chosen from the Emmett study to simulate the Steenland et al. data was 1.5 ppb as this concentration gave the best fit to the data when a 3.8 year half-life was assumed.

To reduce exposure to PFOA, activated carbon filters were installed in Little Hocking and the surrounding communities in 2007. Pre- and post-filtration serum PFOA concentrations were

measured from a cross-section (divided into public water users and bottled water users) of the Little Hocking and Lubeck populations by Bartell et al. (2010). Pre-filtration concentrations were measured in May/June 2007; filtration systems were installed in Lubeck in June 2007 and in Little Hocking in November 2007. Post-filtration concentrations were measured in May/June 2008. Although the population size for this study was small ($N = 132$ for Lubeck; $N = 40$ for LH), the model was used to simulate the pre- and post-filtration serum PFOA concentrations. When simulating the LH population, the pre-filtration drinking water concentration was set at 3.55 ppb; for the Lubeck population the drinking water PFOA concentration ranged from 0.41 to 1 ppb (Bartell et al., 2010), and for the simulations, 0.7 ppb (middle of the range) was used.

Another study performed by Hölzer et al. examined the serum PFOA concentrations in a population in Arnsberg, Germany in July 2006 (Hölzer et al., 2008). As in Little Hocking, the drinking water had been contaminated with PFOA, and this is thought to have arisen from farmers dumping soil conditioner mixed with industrial waste into nearby rivers. The average concentration of PFOA in the German public water was found to be approximately seven times lower than that measured in Little Hocking public water (0.519 ppb versus 3.55 ppb) (Hölzer et al., 2008). The data in the Hölzer et al. study was from approximately 355 people ($n = 101$ for men, $n = 164$ for women, $n = 90$ for kids); as in the Emmett et al. study, only a mean and standard deviation for the population surveyed was available, and it is not known how long the people had been exposed to PFOA; however, it is thought to be only a few years in contrast to the Little Hocking population that has been exposed for over 30 years (Hölzer et al., 2008). After the study was conducted in July 2006, a filtration system was installed to remove the PFOA in the drinking water; the PFOA concentrations in the drinking water were reduced to concentrations below the detection limit (LOD was 10 ng/L). Hölzer et al. measured the serum PFOA concentrations again in July 2007 to quantify the changes in serum concentration (Hölzer et al., 2009). 288 participants from the original 2006 study were included in the 2007 study. In addition to Little Hocking, we used the human model to simulate the serum PFOA concentrations in Arnsberg in both 2006 and 2007.

The half-lives of PFOA and PFOS are much longer in the human than in laboratory animals; the half-lives in humans are reported to be on the order of several years (Andersen et al., 2008; Olsen et al., 2007). Two studies have been performed to investigate the half-life of PFOA, and the gender difference in elimination of PFOA in the rat has not been observed in humans (Olsen et al., 2007). The first, reported by Olsen et al. (2007) examined the half-lives of several perfluorinated chemicals, including PFOA and PFOS, in 26 retired fluorochemical workers. The average serum half-life estimated for PFOA was 3.8 years and that for PFOS was 5.4 years. The second study was performed by Bartell et al. (2010) using the pre- and post-filtration serum PFOA concentration of 200 residents of Little Hocking and the nearby town of Lubeck. The average serum half-life of PFOA estimated from their study was 2.3 years. We considered both half-life estimates when running the model simulations.

PFOA and PFOS have been detected in the serum of the US general population; however, the chemicals are present in much lower levels than what has been found in workers and people drinking the contaminated water. Exposure pathways for the general population are not well characterized (Andersen et al., 2008; Calafat et al., 2007a; Paustenbach et al., 2007). Calafat et al. reported and compared the serum concentrations of several perfluorinated chemicals of the general population from 1999–2000 and 2003–2004 National Health and Nutrition Examination Survey (NHANES) (Calafat et al., 2007b). Due to reduction in production and manufacturing emissions of these chemicals by 2002, the serum concen-

trations of PFOA and PFOS were found to have decreased in the general population (Calafat et al., 2007b). We used the human model to estimate the exposures of the general population to PFOA and PFOS from both 1999–2000 and 2003–2004.

Olsen et al. also examined the decline in serum PFOA and PFOS concentrations (Olsen et al., 2003a, 2008). In these studies, the serum of Red Cross adult blood donors in six geographic locations (100 donors from each location) around the US was analyzed for PFOA, PFOS, and five other fluorochemicals. The first study was conducted in 2000–2001; the follow-up study was performed in 2006. As with the NHANES participants, a decrease in the levels of PFOA and PFOS was observed in the serum of the Red Cross donors. The human model was used to estimate the exposure of all Red Cross donors to PFOA and PFOS in 2000–2001 and in 2006.

2.6. Model structure – human

The PBPK model structure for PFOA and PFOS in the human is the same as that for the monkey (Fig. 2); extrapolation to the human was the principal reason for development of the monkey model. As mentioned in the methods section for the monkey model, the skin compartment was built in because dermal uptake is of interest for the human, and the fat compartment serves as a restriction volume; a larger fat compartment would result in a smaller volume of distribution for these chemicals. Coding and simulations for both the PFOA and PFOS human models was performed in the Berkeley–Madonna program (Macey et al., 2000). Model code can be found in the Supplementary Material.

2.7. Model parameterization – human

2.7.1. Chemical-specific parameters for PFOA and PFOS

Chemical-specific parameters used for the human PFOA model are presented in Table 1. The tissue-plasma partition coefficients used for both chemicals were the same as those used for the monkey. Fiserova-Bergerova (1975) examined partition coefficients for two compounds (Forane and methylene chloride) in several species (man, monkey, dog, and rat). Tissue:blood partition coefficients were very similar for the tissues examined across species; thus, it is reasonable to use the same partition coefficients in both the human and monkey models. For PFOA, the PCs were estimated from tissue concentration data reported by Kudo et al. (2007) for a single IV dose of PFOA to male Wistar rats. For PFOS, PCs were estimated from tissue concentration data from male C57Bl/6 mice (DePierre, 2009). Only the free concentration of chemical is assumed to be available to partition into tissues. As in the monkey and rat, PFOA and PFOS are highly bound to plasma proteins in the human. The species differences in half-lives of these compounds does not appear to be due to differences in protein binding, (Han et al., 2003) so we used the same free fraction of chemical (Free) estimated for the monkeys for PFOA and PFOS in the human models. Urinary elimination rate constants were scaled as $BW^{-0.25}$ and estimated by fitting the model to the data of Harada et al. (2005), where serum and urine PFOA and PFOS concentrations were measured in residents in Kyoto City, Japan.

Similar to the monkey, humans have long half-lives for these chemicals; however, instead of months as for the monkey, the human plasma half-life is on the order of several years. Based on a study of retired fluorochemical workers, the estimated arithmetic mean serum elimination half-life in the human is approximately 3.8 years for PFOA (geometric mean was 3.5 years) and 5.4 years for PFOS (geometric mean was 4.8 years) (Olsen et al., 2007). A more recent study conducted on a population that had previous exposure to PFOA in drinking water estimated a plasma half-life for PFOA of 2.3 years (Bartell et al., 2010). The long half-lives across species (monkey to human) suggests that the same saturable renal

resorption process that is hypothesized to be responsible for the kinetics and long half-life in the monkey is also responsible for the persistence in humans (Andersen et al., 2006). If this is confirmed and the transporter protein(s) that is/are responsible for renal resorption of these chemicals is/are identified, information on the activity of that transporter can then be used to identify susceptible sub-populations due to polymorphic variation. The human PBPK model thus includes the description of renal resorption that we included in the monkey to allow for eventual incorporation of data describing human population variability in renal transporter activities. We assumed that the transporter protein would be the same one in the monkey and the human; studies on the cloning and function of renal transporters in the monkey suggest that the monkey is a good predictor of renal uptake of organic anions in the human (Tahara et al., 2005). Therefore, the transporter affinity constant (Kt) is the same in the human as was estimated for the monkey. We then adjusted the transporter capacity (Tm) for each chemical in the human to get the correct plasma half-lives. Since the clearance of both chemicals is so much slower in the human, the transporter capacity is greater than in the monkeys.

2.7.2. Physiological parameters for the human model – PFOA and PFOS

The physiological parameters used for the human models can be found in Table 2. Body weight, cardiac output, fractional tissue volumes, and fractional blood flows to tissue were the values reported by Brown et al. (1997). Tissue volumes were scaled linearly to body weight and blood flows were scaled as $BW^{0.75}$. To adjust for plasma flows to tissues, tissue blood flows were multiplied by 1-hematocrit.

3. Results

3.1. Monkey model for PFOA

3.1.1. IV data

The simulated plasma and urine PFOA concentrations for a single IV dose of 10 mg/kg PFOA to the monkey were in good agreement with the experimental data (Fig. 3). There appears to be some variability in the plasma clearance of PFOA for the monkeys; at about 60 days post dose, some monkeys have eliminated most of the chemical while some still have higher plasma concentrations. Adjustment of the transporter maximum (Tm) will affect how slowly or quickly the chemical is eliminated in the model. The Tm estimated to the average plasma concentration provided the best fit to the data.

3.1.2. Oral data

As mentioned in Section 2, only the data from the 10 mg/kg daily oral dose of PFOA to the monkeys was used for model development. The simulated plasma and urine concentrations are in good agreement with the observed data from the 10 mg/kg dose (Fig. 4). The model simulates the fast rise to steady state in the plasma and the slower terminal half-life. The model was used to simulate the 30/20 mg/kg daily dose, and the irregular dosing regimen reported in the study was used. However, the observed experimental data for the 30/20 mg/kg dose produced different shapes of plasma and urine concentration curves from the 10 mg/kg dose during the elimination phase. Tm could be increased to fit the high dose data (data not shown); however, because the monkeys in the high-dose group exhibited toxicity, the data observed may be due other effects, such as impaired renal function (decreasing the rate of flow to the filtrate compartment also allowed for fitting of the high dose data). The lower dose is also more relevant for development of the human model, so we chose to use the parameters that best fit the 10 mg/kg dose.

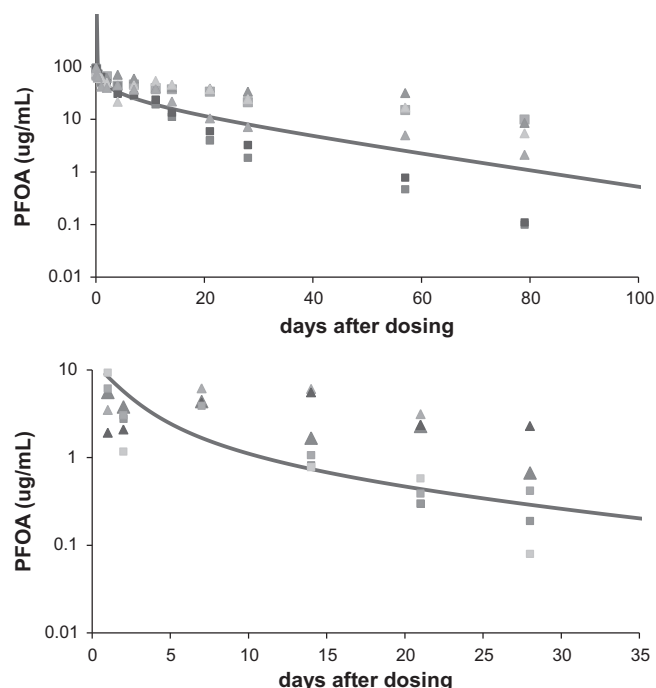


Fig. 3. Comparison of model simulations (lines) with experimental data (points) from Butenhoff et al. (2004). The plots contain PFOA plasma (top) and urine (bottom) concentrations in monkeys administered a single 10 mg/kg iv dose of PFOA and then allowed to recover. Females in triangles; males in squares.

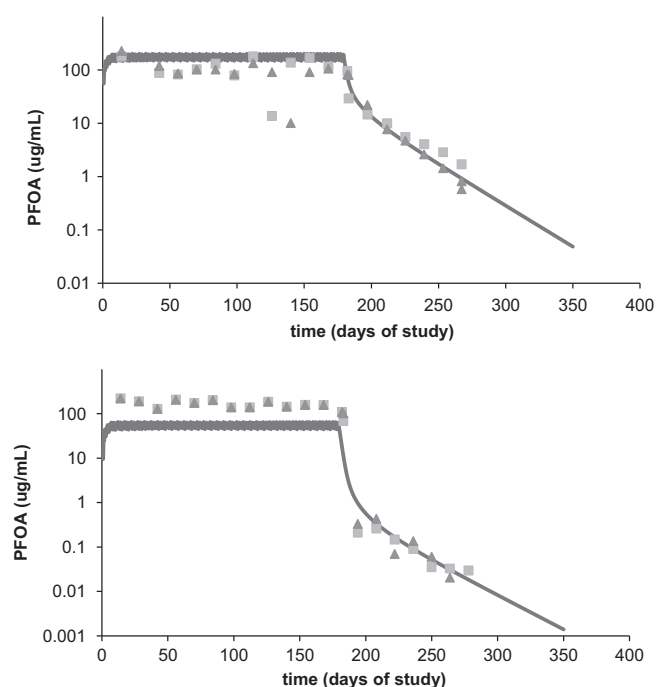


Fig. 4. Comparison of model simulations (lines) with experimental data (points) from Butenhoff et al. (2004). The plots contain PFOA plasma (top) and urine (bottom) concentrations in male monkeys administered daily gavage doses of 10 mg/kg PFOA for 26 weeks and were then allowed to recover.

3.2. Monkey model for PFOS

3.2.1. IV data

The simulated plasma and urine PFOS concentrations for a single IV dose of 2 mg/kg PFOS to monkeys exhibits good

correspondence with the observed data of Noker and Gorman (Fig. 5). Up to about 70 days post dose the monkeys exhibit some variability in plasma elimination of PFOS, but at the last two time points, the animals have similar plasma PFOS levels. Because PFOS appears in the urine at a slower rate than it disappears from the plasma, a storage compartment was added to the model. This “delay” compartment was also used in the compartmental model of Andersen et al. (2006) and Tan et al. (2008) and its use allows for a better fit to the urine data. The biologically based model by Tan et al. (2008) for PFOS in the monkey incorporated a time-dependent description for the free fraction of chemical in the plasma (this was necessary to describe the kinetics resulting from the IV dose). This description was not necessary to implement in the PBPK model in order to describe PFOS pharmacokinetics; thus, the free fraction (Free) was kept constant.

3.2.2. Oral data

The simulated PFOS plasma concentrations resulting from an oral dose of 0.03, 0.15, or 0.75 mg/kg daily to the monkeys are shown in Fig. 6. Overall, the simulated concentrations are in good agreement with the experimental data of Seacat et al. (2002). The 0.15 mg/kg dose lines up well with the experimental data while the highest dose is slightly overestimated by the model, especially during the dosing period. The model simulated a sharper rise in plasma concentration and faster approach to steady state during the dosing period that what was observed in the experimental data. Adjustment of T_m and K_t allowed for better fitting of the high dose curve; however, the best overall fit for all three doses was with the T_m and K_t for the 0.15 mg/kg dose.

3.3. Sensitivity analysis

The normalized sensitivity coefficients for the model parameters for PFOA and PFOS in the monkey with respect to serum PFOA

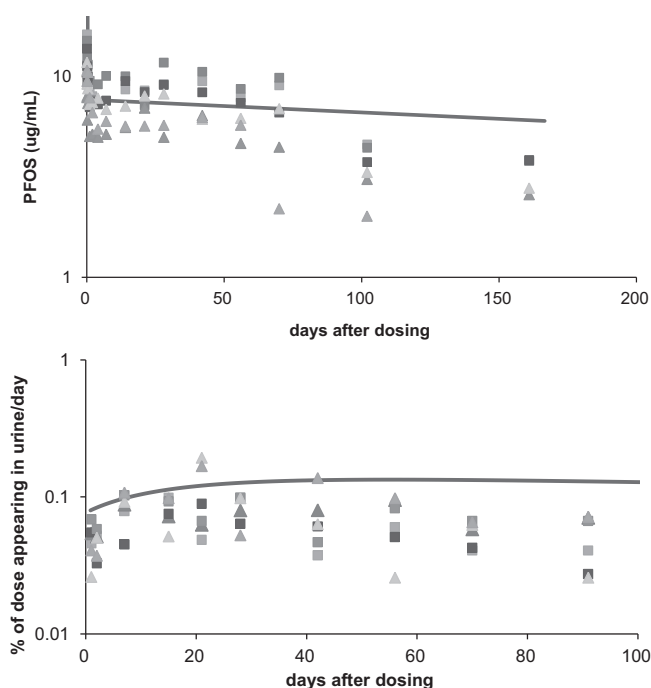


Fig. 5. Comparison of model simulations (lines) with experimental data of Noker and Gorman (2003). The plots contain PFOS plasma (top) and urine (bottom) concentrations after a single iv dose of 2 mg/kg PFOS. Male and female monkeys were administered an iv dose of PFOA and allowed to recover for 5 months. Squares-males, triangles-females.

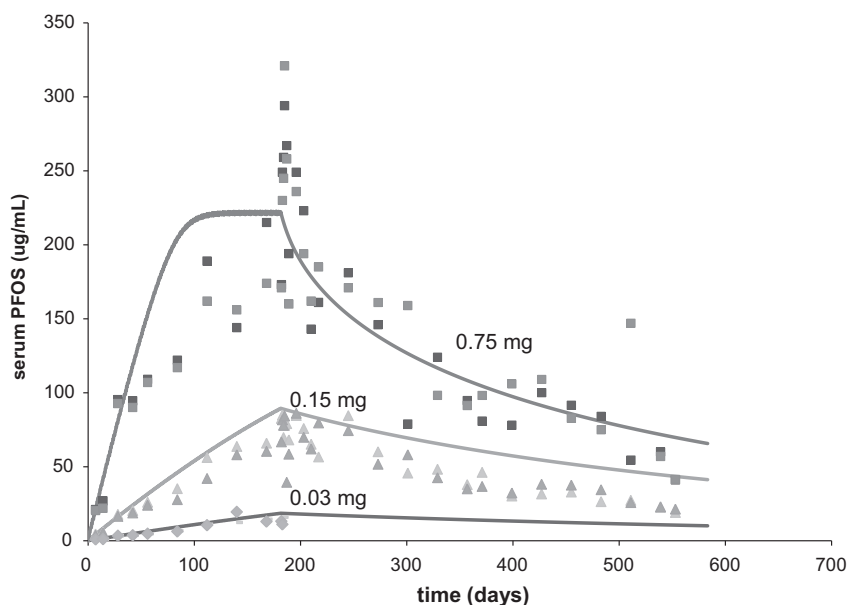


Fig. 6. PFOS in monkey plasma during and after oral dosing for 6 months. Male and female monkeys were administered 0.03, 0.15, or 0.75 mg/kg PFOS orally for 6 months and monitored for a year after dosing stopped.

or PFOS AUC are shown in Fig. 7. Only parameters with sensitivity coefficients greater than 0.1 are shown. For PFOA, sensitivity analysis was performed at 3 and 10 mg/kg daily doses. There do not appear to be any dose-dependent differences in model sensitivity to

any of the parameters shown. Serum PFOA is primarily dependent on the free fraction of chemical in the plasma and the parameters governing elimination, which are the flow to the filtrate compartment and the transporter maximum. Cardiac output and hematocrit will both influence the plasma flow to the filtrate compartment since blood flows are dependent on cardiac output and also must be adjusted for plasma flows because PFOA does not partition into red blood cells. For PFOS, sensitivity analysis was performed at 0.03 and 0.75 mg/kg daily doses. Unlike PFOA, there appear to be some dose-dependent differences in model sensitivity at the low and high doses of PFOS. At 0.03 mg/kg/day, serum PFOS appears to be more dependent on liver volume, liver:plasma partitioning, and rest of body:plasma partitioning than at the higher dose. At 0.75 mg/kg, the free fraction of PFOS in plasma and the elimination parameters (QF_{ilc}, T_{mc}) appear to be more important than at 0.03 mg/kg. At the higher dose, uptake into tissues may be limited and clearance parameters show a greater influence on serum PFOS.

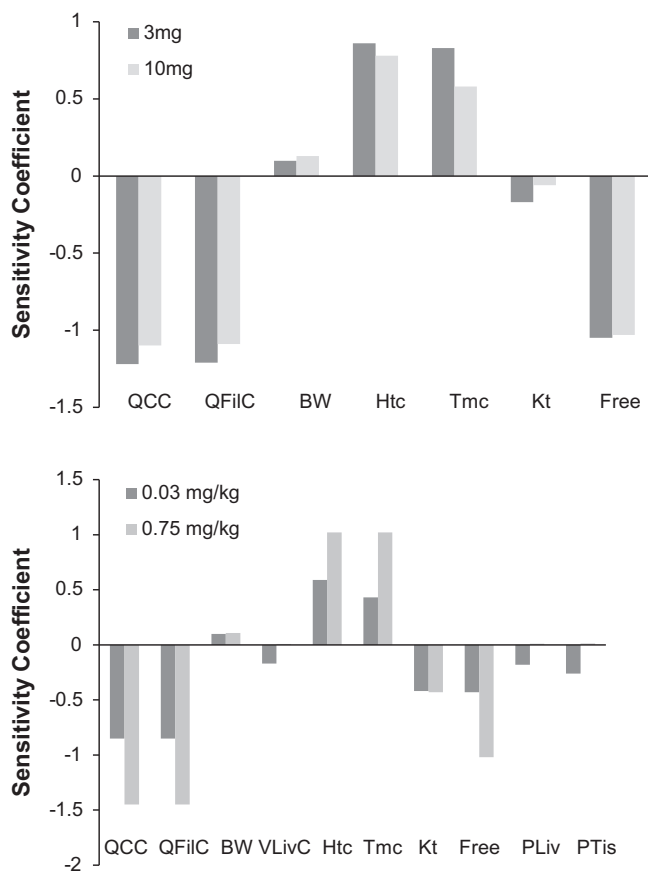


Fig. 7. Calculated sensitivity coefficients for PBPK model parameters for PFOA (top) and PFOS (bottom) in monkeys with respect to serum PFOA or PFOS AUC. Only parameters with SCs > 0.1 are shown on the plots.

3.4. Human model

3.4.1. Comparison with available data

For PFOA, we used the human model to simulate plasma concentrations of the chemical resulting from exposure to contaminated drinking water in both the Little Hocking, Ohio and Arnsberg, Germany populations. The adequacy of the human PFOA model and the corresponding parameters were assessed based on whether the model was able to describe the available plasma concentration data on PFOA in humans, which exhibits high variability. Overall, agreement between the experimental data and the model simulations was good. Further efforts to refine the model parameters will require more detailed kinetic studies in the human. The exposure conditions (drinking water rate, concentration of PFOA on drinking water) used to simulate the different populations are in Table 3.

3.4.1.1. Little Hocking, Ohio. Since it is not known how long the people had been exposed (i.e., drinking and using the water), the model was run until the plasma PFOA concentration reached steady state and until the model predictions were within the measured

Table 3

Exposure estimates used in simulations with human PBPK model for PFOA.

Parameters	Values	Sources
PFOA in water	Mean: 3.55 ppb (Little Hocking)Mean: 0.519 ppb(Arnsberg)	Emmett et al. (2006) and Hölzer et al. (2008)
Intake	Mean: 11 mL/kg/day(range: 1–95)	EPA-822-R-00-008
PFOA in serum (Little Hocking)(N = 291)	Mean: 448 µg/LMedian: 374 µg/Linterquartile range:221–576 µg/L	Emmett et al. (2006)
PFOA in serum (Arnsberg, Germany)(N = 101; men only)	Mean: 28.5 µg/LSD: 12.9	Hölzer et al. (2008)

mean and standard deviation. For the Little Hocking population, the model had to be run for approximately 30 years before steady state and the mean PFOA plasma concentration was reached. In addition to making sure the model reached steady state and the mean/SD, we made sure that we had adjusted the transporter maximum (T_m) to yield the proper half-life. When simulating the Little Hocking population, we assumed that their only exposure to PFOA was through the drinking water as this was their primary exposure source. Because there are two different plasma half-life estimates that we found in the literature, (Bartell et al., 2010; Olsen et al., 2007) the model was run assuming both half-lives. Fig. 8 shows the plot of simulated plasma concentrations with the measured data from Emmett's study. From the plot, it can be seen that either half-life (3.8 versus 2.3 years) is possible; both yield simulated concentrations that are within the mean and SD. The Little Hocking Water Association (LHWA) website contains plasma PFOA concentration data for 25 residents. The average and standard deviation calculated from this data were slightly different that reported in Emmett's 2006 study, so for comparison with the simulations, both data sets (mean and standard deviations for LHWA website data (triangles) and mean and interquartile range for Emmett data (squares)) are shown in Fig. 8.

In addition to the data from Emmett et al. and the LHWA website, data collected from the participants in the C8 Health Project was also simulated with the human model. The purpose of the C8 Health Project was to assess the serum PFOA levels of the LH and surrounding area residents that were served with water containing PFOA. This data was collected from August 2005–July 2006 and was published by Steenland et al. (2009). As with the Emmett data, the model was run for 30 years of exposure in order to

reach steady state, and both available estimates of the plasma half-life for PFOA were tested (2.3 versus 3.8 years). Fig. 9 shows the plot of the simulated plasma concentrations along with the measured data from both Emmett's study and the C8 Health Project study. For this plot, a drinking water concentration of 3.55 ppb was used. The mean serum PFOA concentration measured from the LH residents participating in the C8 Health Project is ~227 ppb, which is approximately half that measured in the Emmett et al. study (~448 ppb); however, the mean value from the C8 Health Project overlaps with the lower end of the interquartile range (221–576 ppb) measured in Emmett's study.

The serum half-life of PFOA is not the only factor that probably has variability for this population. No drinking water concentrations of PFOA were listed in the Steenland et al. study; however, in the Emmett et al. study, the measured drinking water concentrations ranged from 1.5 to 7.2 ppb (mean = 3.55 ppb). In Fig. 9, the 3.8 year half-life appeared to overestimate the serum PFOA concentrations measured from the C8 Health Project participants, so the model was then run with a reduced drinking water concentration (1.5 ppb) but the half-life was kept at 3.8 years. Fig. 10 shows the results of this (bottom) and also the model predictions plotted with the Emmett et al. data (top) and the 3.55 ppb intake. Lowering the drinking water concentration allowed for good predictions with the 3.8 year half-life. The people in the Emmett cohort were from LH only, which was the area with the highest water concentration of PFOA, while the participants in the C8 Health Project were from LH and surrounding communities where the water concentration of PFOA was lower; thus, it is reasonable that the people from the C8 Health Project have lower serum PFOA concentrations. In addition to variability in serum half-life and the concentration of

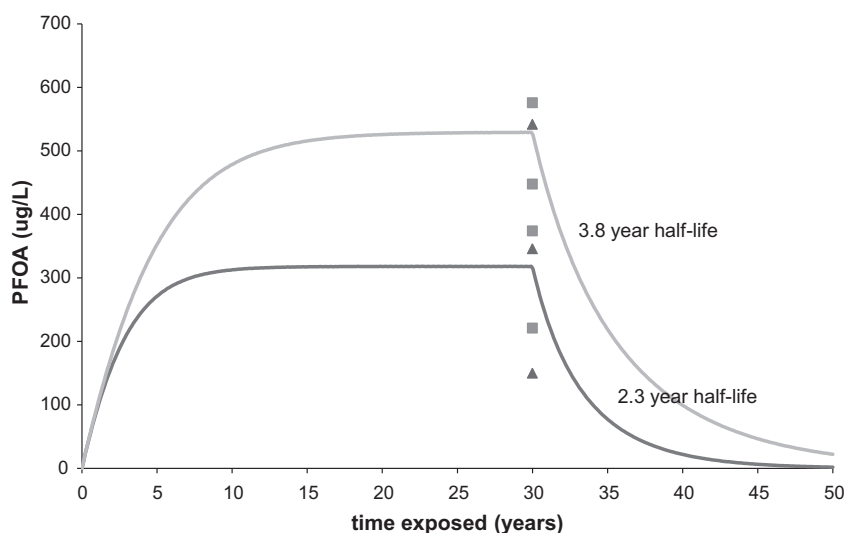


Fig. 8. Comparison of model simulations (lines) with experimental data from Emmett et al. (2006) (squares) and the Little Hocking Water Association website in 2005 (triangles). The plot contains PFOA plasma concentrations for the Little Hocking, OH population that was exposed to drinking water contaminated with PFOA (3.55 ppb). The simulations were run for an exposure period of 30 years assuming two different half-lives: 3.8 years (red curve) from a study of occupational workers by Olsen et al. (2007), and 2.3 years (blue curve) from a study by Bartell et al. (2010). Both half-lives appear to be probable.

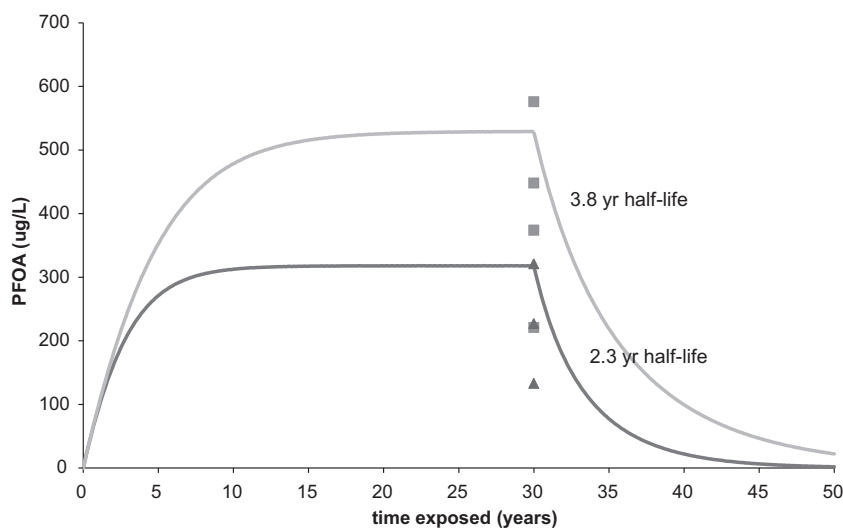


Fig. 9. Comparison of model simulations (lines) with experimental data from Emmett et al. (2006) (squares) and Steenland et al. (2009) (triangles). The plot contains PFOA plasma concentrations for the Little Hocking, OH population that was exposed to drinking water contaminated with PFOA. Both data sets were collected in 2005–2006. The simulations were run for an exposure period of 30 years assuming a half-life of 3.8 years (top curve) and 2.3 years (bottom curve). The data from Steenland et al. (2009) is from the participants in the C8 Health Panel. The mean serum concentration is ~227 ppb, about half of that measured by Emmett et al. in 2006 (mean ~448 ppb). A drinking water concentration of 3.55 ppb PFOA was assumed.

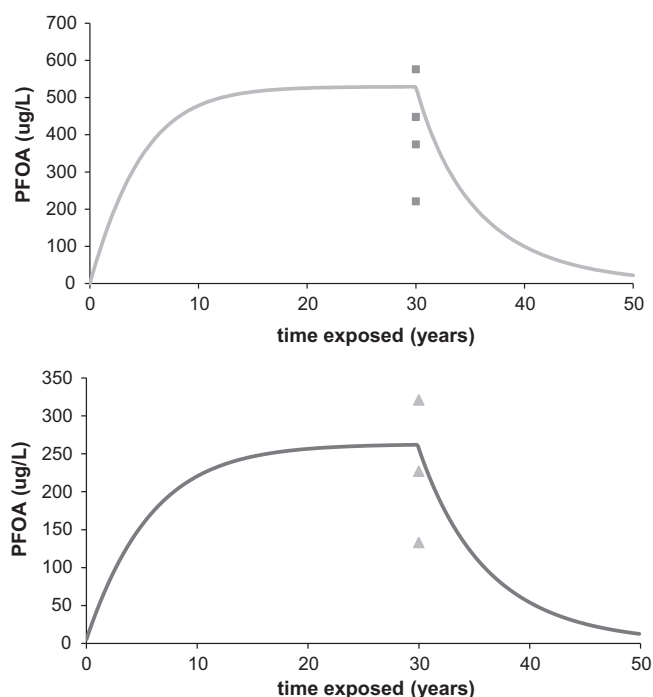


Fig. 10. Comparison of model simulations (lines) with experimental data from Emmett et al. (2006) (top, squares) and Steenland et al. (2009) (bottom, triangles). The plots contain PFOA plasma concentrations for the Little Hocking, OH population that was exposed to drinking water contaminated with PFOA. Both data sets were collected in 2005–2006. The simulations were run for an exposure period of 30 years and a serum half-life of 3.8 years was assumed. The water concentration of PFOA for simulating the Emmett data sets was set at 3.55 ppb; however, for the Steenland data, the drinking water concentration had to be reduced to 1.5 ppb if a 3.8 year half-life is used. The water concentrations in LH between 2002 and 2005 were found to range from 1.5 to 7.2 ppb (Emmett et al., 2006), so the people studied by Steenland et al. simply may have been in a lower exposure region. Variable water concentrations and/or half-lives of PFOA in individuals is likely.

PFOA in drinking water in various regions, drinking water rates and how much exposure comes from drinking water versus other sources most likely differ among the populations examined as well.

To reduce exposure to PFOA in the drinking water, an activated carbon filtration system was installed in Little Hocking and the nearby community of Lubeck in 2007 (Bartell et al., 2010). Pre- and post-filtration serum PFOA serum concentrations were measured by Bartell et al. in a cross-section of the populations, and the participants were divided into public water or bottled water users (Bartell et al., 2010). The model was used to simulate the serum PFOA concentrations of the public water users in both Lubeck and LH (Fig. 11). For both Lubeck and LH, both available serum half-lives of PFOA were taken into consideration when running the simulations. For LH public water users (top), exposure was assumed to be 30 years (to reach steady-state) plus 6 months, as filtration systems were installed 6 months after the pre-filtration serum concentrations were measured. The pre-filtration water concentration was assumed to be 3.55 ppb; when filtration started, the exposure was simply shut off, as the PFOA water concentration after the filtration system was installed was nonquantifiable or nondetectable (Bartell et al., 2010). For Lubeck public water users (bottom), exposure was assumed to be 30 years and then it was shut off; filtration systems were installed in Lubeck right after the pre-filtration serum concentrations were measured. PFOA in drinking water samples in Lubeck ranged from 0.41 to 1 ppb between 2001 and 2006; 0.7 ppb (middle of this range) was used in running the simulations. Due to the small population sizes used in the Bartell et al. study, the $\pm$ SD from the mean for the post-filtration serum PFOA concentrations are quite small. However, from the pre-filtration concentrations, it appears from Fig. 11 that either the 2.3 or 3.8 year half-life is possible. A larger population size would be helpful in examining the extent of the variability in serum PFOA concentration.

3.4.1.2. Arnsberg, Germany. As with the people of Little Hocking, it is not known how long the people of Arnsberg were exposed to PFOA in their drinking water. It is believed that the contamination occurred in the time period between 2000 and 2006; this population was therefore not exposed as long as the people of Little Hocking (Hölzer et al., 2008). The concentration of PFOA in the drinking water was approximately 7 times lower than that measured in Little Hocking (0.519 ppb versus 3.55 ppb). For the Arnsberg population, the plasma PFOA concentration reached measured mean

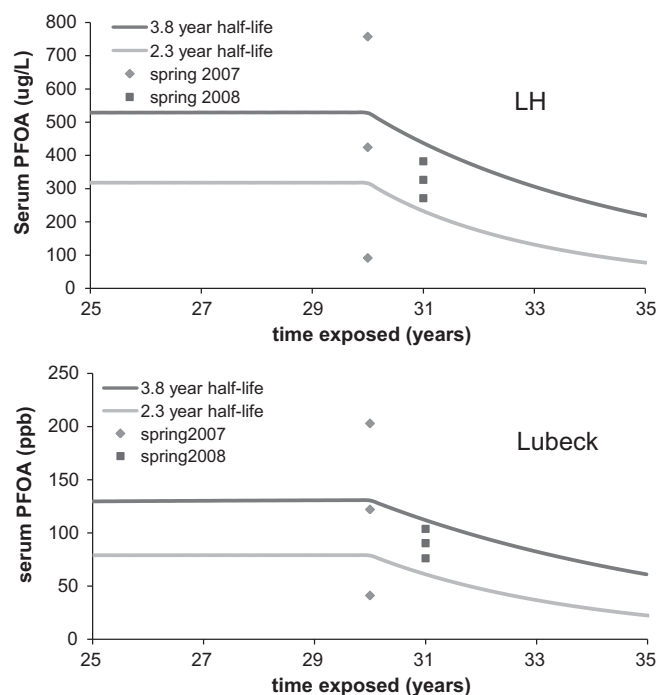


Fig. 11. Comparison of model simulations (lines) with experimental data from Bartell et al. (2010) (points). The plots contain PFOA plasma concentrations for the Little Hocking (top) and Lubeck (bottom), OH populations that were exposed to drinking water contaminated with PFOA. Serum PFOA was measured in spring 2007 (before filtration systems were installed; diamonds) and in spring 2008 (~6 months after filtration started; squares). The simulations were run for an exposure period of 30 years assuming a half-life of 3.8 years (top curves) and 2.3 years (bottom curves). Average drinking water concentration of PFOA in LH was 3.55 ppb and 0.7 ppb in Lubeck. The population size for this study was small ($N = 132$ for Lubeck; $N = 40$ for LH); however, both half-lives appear to be likely.

within a few years (Fig. 12); however, the plasma concentration does not reach steady state in this time. Two studies were performed on this population. The initial study was performed in July 2006, (Hölzer et al., 2008) after which a filtration system was

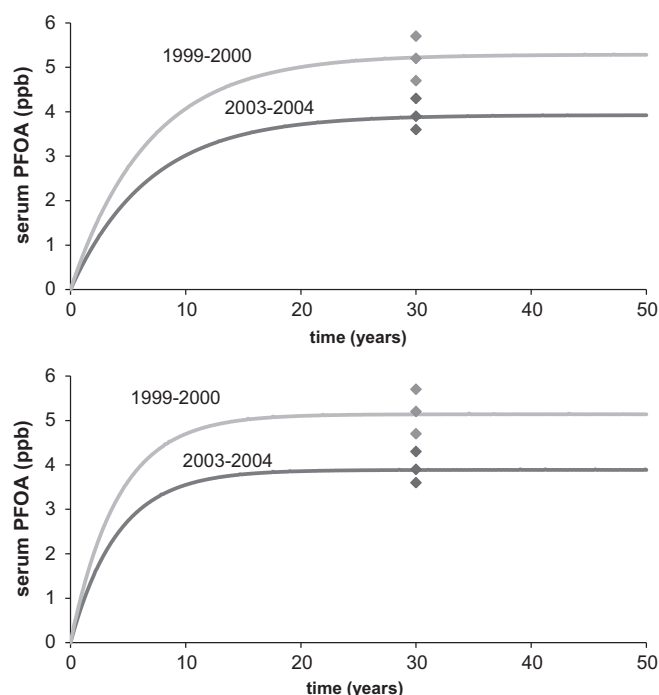


Fig. 13. Comparison of model simulations (lines) with serum PFOA concentration of general population (points). The plots contain the simulated serum PFOA concentrations using a half-life of 2.3 years (top) and 3.8 years (bottom). Estimated general population exposure was simulated as a direct input to blood. For a 2.3 year half-life, the estimated exposure range from 1999 to 2000 was $4.25\text{--}5.2e^{-4}$ ug/kg/day and $3.3\text{--}3.9e^{-4}$ ug/kg/day for 2003–2004. For a 3.8 year half-life, the estimated exposure range was $2.6\text{--}3.2e^{-4}$ ug/kg/day for 1999–2000 and $2.0\text{--}2.4e^{-4}$ ug/kg/day for 2003–2004.

installed by the waterworks. The drinking water concentration was lowered considerably by filtration (mostly below the LOD, which was 10 ng/L). The follow-up study was then performed in July 2007 to examine the effects of the filtration on serum PFOA concentration (Hölzer et al., 2009). The authors of the studies

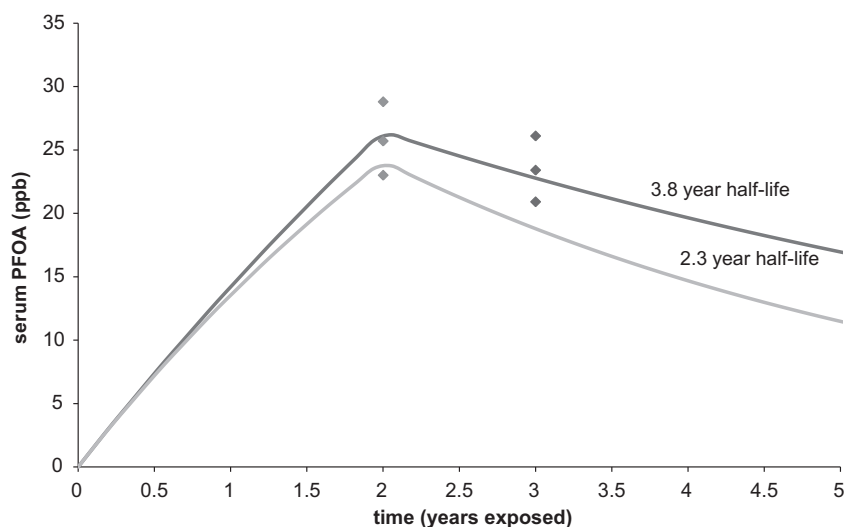


Fig. 12. Comparison of model simulations (lines) with experimental data from Hölzer et al. (2008) and Hölzer et al. (2009) (points). The plot contains PFOA plasma concentrations for a population in Arnsberg, Germany that was exposed to drinking water contaminated with PFOA (0.519 ppb). A filtration system was installed right after the initial serum concentrations were measured (July 2006), and the serum PFOA concentrations were measured a year after this (July 2007). The exposure duration is unknown, but it appears to be on order of a few years. The simulations pictured were run for two years assuming two different half-lives: 3.8 years (blue curve) from Olsen et al. (2007) and 2.3 years (red curve) from Bartell et al. (2010). As for the Little Hocking population, both half-lives appear to be likely.

measured the plasma concentration of PFOA in men, women, and children; however, since the means were similar for all groups (28.5 ppb for men, 26.7 ppb for women, and 24.6 ppb for kids in the initial study; 23.4 ppb for men, 18.8 ppb for women, and 17.4 for kids in the follow-up study), only the simulations for the men are shown in Fig. 12. As with the Little Hocking simulations, we assumed that drinking water was the only source of exposure to PFOA and the simulations were run using both a 3.8 and 2.3 year half-life. From Fig. 12, it can be seen that either half-life fits with the data (although the 3.8 year half-life yields a slightly better fit).

3.4.1.3. General population. PFOA and PFOS have also been measured in the blood of the general population (Calafat et al., 2007a; Calafat et al., 2007b). These levels are much lower than those found in the serum of the residents of Little Hocking and Arnsberg; however, the exposure sources are unknown and could potentially be from a number of sources, including drinking water, food and food packaging, consumer products, and in the air. Serum concentrations of PFOA and PFOS from the general population are available from 1999–2000 to 2003–2004 from the National Health and Nutrition Examination Survey (NHANES). We used the model to simulate the serum concentrations and to estimate how much PFOA and PFOS the general population might be exposed to daily. Figs. 13 and 14b show the comparison of the model simulations with the measured concentrations reported by Calafat et al. (2007b). The top plot in Fig. 13 shows the simulation for PFOA assuming a 2.3 year half-life and the bottom plot assumes a 3.8 year half-life. Fig. 14b shows the simulation for PFOS assuming a 5.4 year average serum elimination half-life (Olsen et al., 2007). Because the exposure sources of PFOA and PFOS are unknown, the simulations were performed with chemical exposure being a direct input to blood (effectively an IV dose). The major US manufacturer of PFOS and compounds that can degrade to PFOS ceased all production by the end of 2002, and, while PFOA continues to be produced and used in the US, emissions have been reduced. This is apparently reflected in the serum concentrations of both PFOA and PFOS: the geometric mean of the serum PFOA concentration decreased from 5.2 ppb in 1999–2000 to 3.1 ppb in 2003–2004 and from 30.4 to 20.7 for PFOS. As far as exposure estimates go, if a 2.3 year half-life is assumed for PFOA the exposure range is $4.25\text{--}5.2\text{e-}4$ ug/kg/day in 1999–2000 and $3.0\text{--}3.9\text{e-}4$ ug/kg/day in 2003–2004. If a 3.8 year half-life is assumed, the exposure range is $2.6\text{--}3.2\text{e-}4$ ug/kg/day in 1999–2000 and $2.0\text{--}2.15\text{e-}4$ ug/kg/day in 2003–2004. For PFOS, the exposure range is $2.3\text{--}3.0\text{e-}3$ ug/kg/day in 1999–2000 and $1.6\text{--}1.9\text{e-}3$ ug/kg/day in 2003–2004. These ranges were estimated using the 95% confidence intervals for the geometric means reported by Calafat et al. (2007b). For PFOA, if we compare these estimates to what the average person in Little Hocking ingested daily in 2006 (assuming consumption of 11 mL water/kg BW/day and 3.55 ppb PFOA in drinking water), the general population exposure is approximately 195 times less than that of a person in Little Hocking. For PFOS (if the estimate of 0.34 ppb PFOS in drinking water in LH is used; see section on PBPK human model for PFOS), if the general population estimates are compared to that of LH (assuming consumption of 11 mL/kg BW/day of water), the general population exposure is approximately 2.3 times less than for the people in LH.

In addition to the general population data from the NHANES, Olsen et al. reported the serum PFOA and PFOS concentrations in adult Red Cross blood donors from 2000–2001 to 2006 in six geographic locations around the US ($N = \sim 600$ in both studies; ~ 100 samples from each clinic) (Olsen et al., 2003a, 2008). We used the model to simulate the serum concentrations of PFOA and PFOS at both time periods in the donors and also the decline in serum concentrations of these chemicals after the phase-out of PFOS-producing chemicals. In the first set of simulations, exposure to both

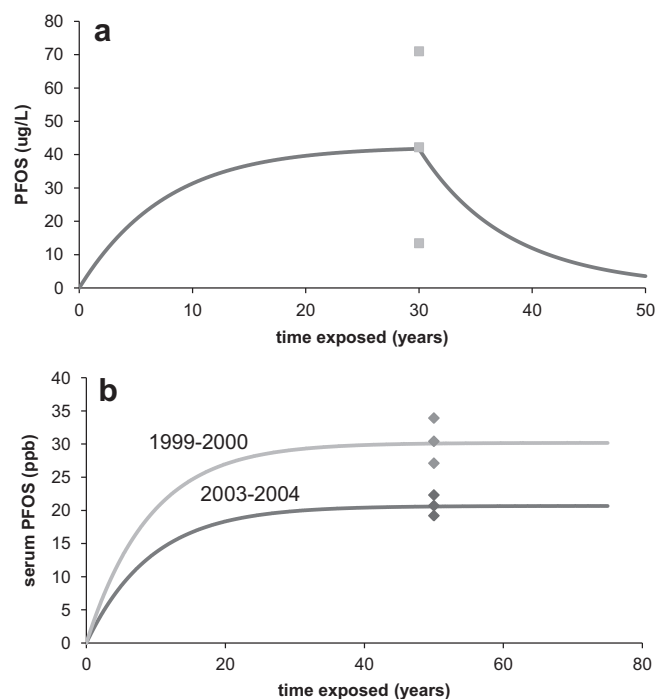


Fig. 14. (a) Comparison of model simulations (lines) of serum PFOS concentration with Little Hocking Water Association website data (squares). The data from the website is the average and SD of the plasma concentrations of residents in Little Hocking ($N = 25$). The concentration of PFOS in the drinking water was not reported (or unknown) at this time. As for PFOA, the simulation for PFOS was run to reach steady state (30 years) and to reach the average concentration measured in plasma (the concentration of PFOS in the drinking water was estimated). A half-life of 5.4 years (from occupational data of Olsen et al.) was assumed for the simulation. (b) Comparison of model simulations (lines) with serum PFOS concentration of general population (points). The plots contain the simulated serum PFOS concentrations using a half-life of 5.4 years. Estimated general population exposure was simulated as a direct input to blood. The estimated exposure range was $2.3\text{--}3.0\text{e-}3$ ug/kg/day for 1999–2000 and $1.6\text{--}1.9\text{e-}3$ ug/kg/day for 2003–2004.

chemicals was assumed to be constant (the simulation was run so that concentrations reached steady state, but exposure was not stopped; Figs. 15 and 16, top panels). For PFOS, the half-life was assumed to be that estimated from retired fluorochemical workers (arithmetic mean of 5.4 years) by Olsen et al., for PFOA, both the 3.8 year arithmetic mean half-life estimated by Olsen et al., and 2.3 year half-life estimated by Bartell et al. were considered (Bartell et al., 2010; Olsen et al., 2007). The estimated daily exposure to PFOS in 2000–2001 was $2.2\text{--}4.5\text{e-}3$ ug/kg BW; in 2006, it was estimated between $8.5\text{e-}4$ and $1.8\text{e-}3$ ug/kg BW. The estimated daily exposures for PFOA in 2000–2001 using the model are $3.2\text{--}6.0\text{e-}4$ ug/kg BW (2.3 yr half-life) and $1.9\text{--}3.7\text{e-}4$ ug/kg BW (3.8 yr half-life); the estimated daily exposures in 2006 are $1.3\text{--}2.65\text{e-}4$ (2.3 year half-life) and $2.2\text{--}4.4\text{e-}4$ (3.8 year half-life). These estimates were made using the 95% confidence intervals for the geometric means and also the interquartile ranges reported by Olsen et al. (2003a, 2008) and are close to the values estimated for the participants in the NHANES study, which ranged from $1.6\text{e-}3$ to $3.0\text{e-}3$ ug/kg/day for PFOS and $3.0\text{--}5.2\text{e-}4$ ug/kg/day for PFOA. In the second set of simulations, the concentrations were allowed to reach steady state and then exposure was shut off two years after the initial measurements were taken. We assumed that steady state was reached when the initial measurements were taken (2000–2001); 3 M completely ceased production of PFOS-producing chemicals in 2002. For both PFOA and PFOS, the exposure was estimated for all donors in the study in both 2000 and 2006. The estimated exposure in 2000 was then used to examine

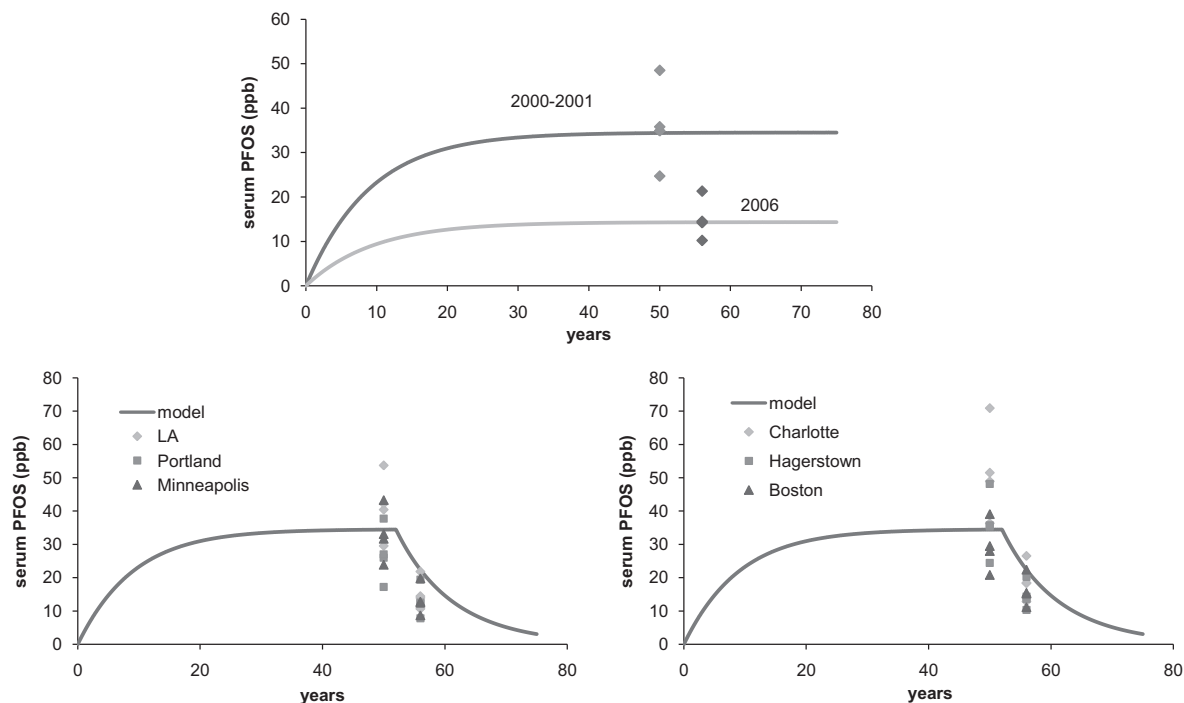


Fig. 15. Comparison of model simulations (lines) with serum PFOS concentration of Red Cross adult blood donors in 2000 and 2006 (points). A serum half-life of 5.4 years for PFOS was assumed. The top panel shows the serum PFOS concentrations from donors in all six locations examined, assuming constant exposure in both 2000 (estimated exposure was 0.0022–0.0045 ug/kg) and 2006 (estimated) exposure was 0.00085–0.0118 ug/kg). The bottom panel shows the serum PFOS concentrations with exposure shut off 2 years after the initial measurements were taken to simulate the phase-out of PFOS-producing chemicals. The points are the serum PFOS concentrations measured in each of the six cities. The exposure estimated for all donors and the 5.4-year half-life yield good agreement for the individual cities.

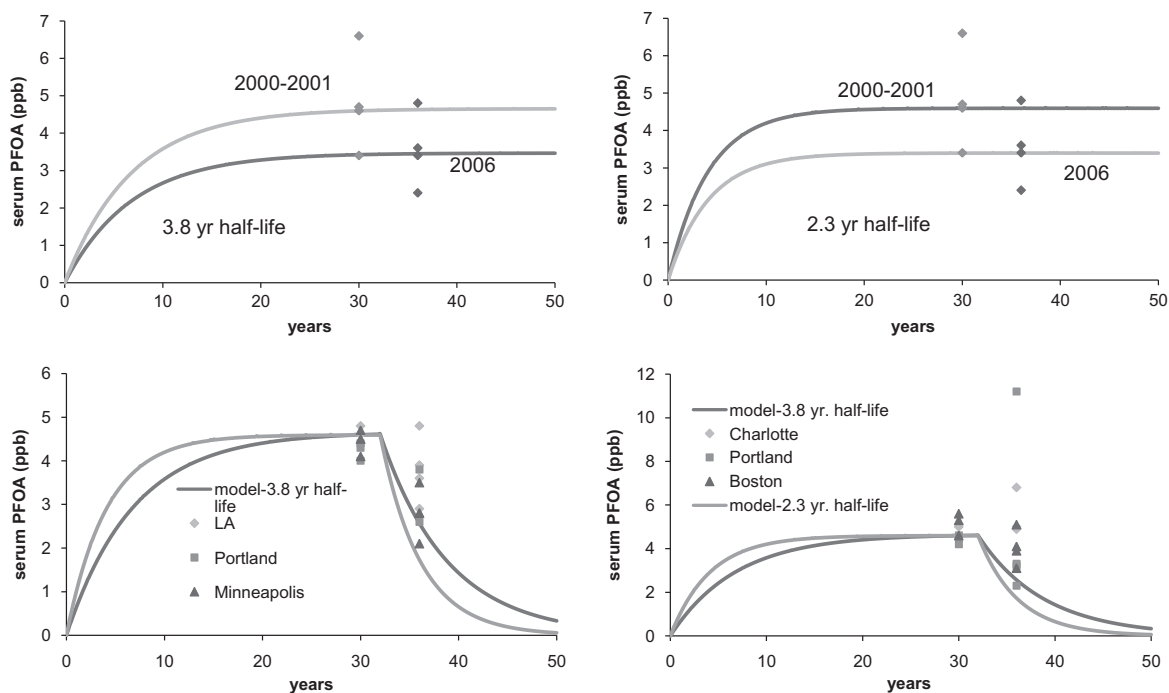


Fig. 16. Comparison of model simulations (lines) with serum PFOA concentration of Red Cross adult blood donors in 2000 and 2006 (points). A serum half-life of 3.8 and 2.3 years for PFOA was assumed. The top panel shows the serum PFOA concentrations from donors in all six locations examined, assuming constant exposure in both 2000 and 2006. The bottom panel shows the serum PFOA concentrations with exposure shut off 2 years after the initial measurements were taken to simulate the phase-out of perfluoroalkyl compounds. The points are the serum PFOA concentrations measured in each of the six cities. The exposure estimates for all donors and both half-lives yield good agreement for the individual cities.

the decline in serum concentration that was observed in 2006 for each of the six cities in the study (Figs. 15 and 16, bottom panels).

Although the concentration ranges for the donors in each city may vary, the estimated exposure for all donors in 2000 appears to yield

good model predictions for each city and for the decrease in concentration observed in 2006.

3.4.1.4. PBPK human model for PFOS. Data in the human is extremely limited for PFOS. The largest US manufacturer of PFOS completed its phaseout of the chemical between 2000 and 2002 (Butenhoff et al., 2006). As for PFOA, no definite association between exposure to PFOS and adverse health effects in humans has been established. Serum PFOS concentrations have been measured in residents from Little Hocking; however, PFOS was present in levels similar to those measured from the general US population (Frisbee et al., 2009; LHWA, 2005). The study conducted by Olsen et al. (2007) with retired fluorochemical workers estimated an average half-life of 5.4 years for PFOS, and the LHWA website included serum PFOS concentrations of 25 individuals. Using this data, we were able to estimate a transporter maximum for the human ($T_{mc} = 3.5$) that would yield the estimated 5.4 year half-life and run the model to reach the measured mean serum PFOS concentration in the Little Hocking people. Fig. 14a shows the comparison of the simulated serum PFOS concentration compared to the measured average and standard deviations from the LHWA website data. No PFOS concentration in the drinking water could be found for Little Hocking; therefore, we had to estimate this to achieve the correct serum concentration. The mean estimated concentration was 0.34 ppb, which is ten times less than the PFOA concentration in the drinking water. The water levels are most likely lower because the nearby production plant did not produce PFOS. As for PFOA, it took about 30 years of exposure to reach steady state and it was assumed that drinking water was the only source of PFOS exposure. Although the existing data for human PK for PFOS is scarce, we were able to make a human model for this chemical that works reasonably well. Further refinement of the model will require that more data for PFOS in humans becomes available.

3.4.1.5. Predicted serum PFOA half-lives. Only two studies have been conducted that have estimated the half-life of PFOA in humans (Bartell et al., 2010; Olsen et al., 2007). The average half-life estimated from the 26 retired fluorochemical workers (3.8 years) was longer compared to the environmentally exposed populations in Lubeck and Little Hocking (2.3 years). From running the human model for the Little Hocking and Arnsberg populations and taking both half-lives into consideration, it was observed that both estimated half-lives appear likely; both allowed for the predicted PFOA serum concentrations to fall within the measured means and standard deviations (Figs. 8–12). In addition to running the model to compare the simulations with available data, we used the model to predict the serum half-lives of the 25 individuals for whom data was provided on the Little Hocking Water Association website. The data on the individuals and their serum PFOA concentration is given in Table 4. For simplicity, the drinking water concentration was assumed to be 3.55 ppb and the drinking water rate was set at 11 mL/kg BW/day for each person unless noted in the table (USEPA, 1997). For men, a standard BW of 70 kg was used; for women, a BW of 58 kg was used in the model. To predict the serum half-life for each person in the table, the exposure period was modified to reflect how long they had been drinking the water (the serum concentration given in the table was assumed to be the maximum serum concentration and the end-of-exposure concentration). To get the correct serum PFOA concentration for each individual, the transporter maximum (T_m) was adjusted. The predicted half-lives can be found in Table 4. The mean half-life of the 25 people is approximately 2.6 years, and they range from 0.8 to 8 years. There were two individuals in the table for whom the maximum serum PFOA concentration could not be reached simply by adjusting T_m . In order to reach their serum concentrations in the time period that they were exposed, their drinking water intake rates had to be increased from 11 mL/kg BW/day to 29 and 22 mL/kg BW/day. These increased drinking water rates are not

Table 4
Model prediction of serum PFOA half-lives in the Little Hocking population.

LHWA	May-05				
Sex	Age	ppb PFOA	Years drinking H ₂ O	T_{mc} (PFOA)	PFOA Half-life (years)
F	46–55	281	28	5.5	2.04
M	46–55	395	51	7.5	2.89
F	46–55	248	55	4.9	1.82
F	26–35	228	27	4.5	1.67
M	<15	629	3	10	3.7 ^a (29 mL/kg BW/day)
F	>65	442	35	8.7	3.23
M	36–45	220	35	4.1	1.61
F	36–45	112	9	2.2	0.82
M	56–65	532	34	10	3.87
F	26–35	116	4	2.3	0.86
F	26–35	211	6	4.3	1.61
M	<15	268	4	7.1	2.93
F	<15	436	3	10	3.81 ^a (22 mL/kg BW/day)
M	26–35	216	4	4.7	1.86
M	36–45	325	25	6.2	2.4
F	46–55	475	33	9.25	3.44
M	46–55	432	33	8.2	3.15
F	>65	358	25	7	2.64
M	>65	346	20	6.5	2.52
M	36–45	1040	37	20	7.71
F	36–45	240	35	4.7	1.75
M	56–65	204	16	3.8	1.46
M	26–35	176	24	3.3	1.28
F	16–25	488	16	9.7	3.59
M	>65	238	34	4.5	1.74
Mean		346.24	23.84		2.576
SD		196.350087			
Median		281			

^a Indicates that serum PFOA concentration could not be attained by adjusting only T_{mc} . Drinking water rate had to be increased from 11 mL/kg BW/day to the amount indicated. Drinking water concentration was set at 3.55 ppb for each individual.

unreasonable; these individuals may have had higher activity levels (USEPA, 1997), simply drank more water, or other physiological factors could be involved.

The estimated half-lives from retired workers and the environmentally exposed population obviously fit into the predicted range of half-lives from the people on the LHWA website (3.8 and 2.3 years). The retired workers in the study were no longer occupationally exposed, but their serum concentrations were higher than the general population (Calafat et al., 2007b; Olsen et al., 2007). Fig. 17 compares the distributions of the predicted half-lives from Little Hocking with those calculated for the retired workers. The workers' average initial serum concentration was higher than that of the average concentration in Little Hocking (691 ppb for workers versus 346 ppb for LH), but the half-lives for both groups are comparable. Little Hocking has a higher frequency of shorter half-lives (in the 1–2 year range) while the workers have a higher frequency of half-lives in the 3–4 year range. Neither group has many individuals with a half-life longer than 4 years.

Prediction of individual half-lives gives an idea of the variability in how long different people retain or eliminate PFOA. From the half-lives predicted by the model, the half-life does not appear to be dependent on how long the people were exposed to PFOA in the drinking water (Fig. 18, top). There is no correlation between length of exposure and half-life; the person with the longest half-life (8 years) was drinking the water for 37 years, but people who drank the water for over 50 years had shorter half-lives (2–3 years). The same was found for the fluorochemical workers; there does not appear to be a correlation between how long they worked in the plant and their half-life of PFOA (Fig. 18, bottom). If renal resorption is the main process responsible for the long half-lives observed in the human, the variability in elimination of PFOA could be due to polymorphisms in renal transporters or variability in expression of the transporter(s). Once the transporter responsible for retention of PFOA in human is identified, information on the activity and expression of the transporter can then be used to determine if there are potentially sensitive individuals.

4. Discussion

Previously, a biologically motivated compartmental model was developed that was able to describe the PK of PFOA and PFOS in the cynomolgus monkey. However, to extrapolate the model to a human to aid in risk assessment, a physiologically based model with realistic tissue volumes, blood flows, and biochemical properties of PFOA and PFOS were needed. We have therefore developed a PBPK model for PFOA and PFOS in the cynomolgus monkey, and the model is able to describe the available pharmacokinetic data in monkeys exposed to these chemicals via IV and oral dosing.

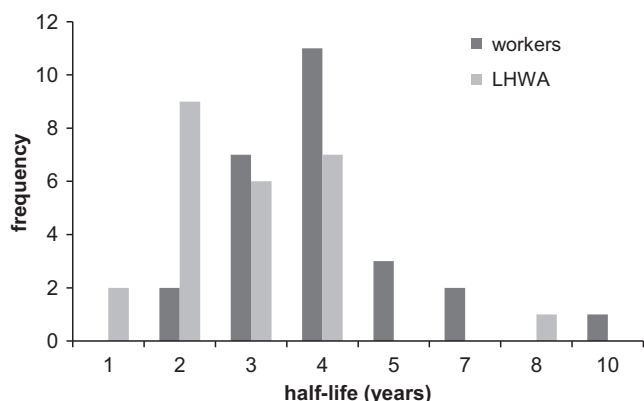


Fig. 17. Distribution of half-lives for LH residents and retired fluorochemical workers.

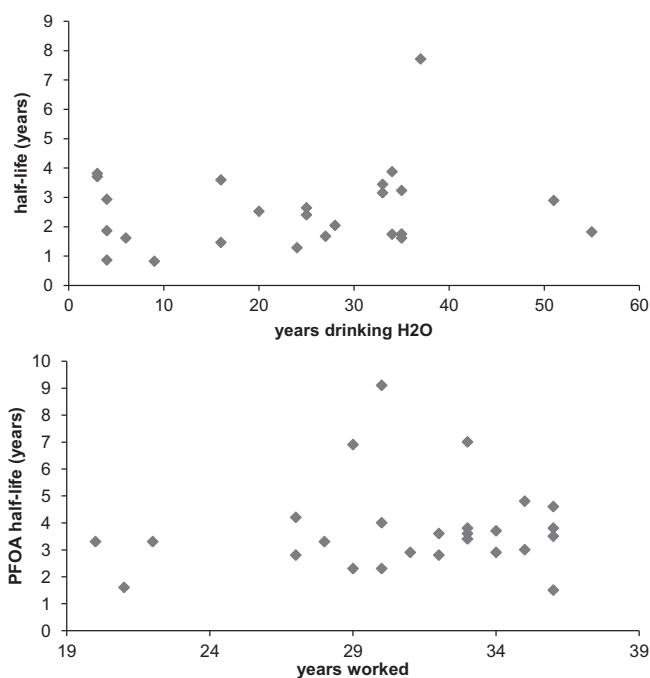


Fig. 18. Half-lives of residents of LH vs. exposure period (top) and retired workers (bottom). There is no correlation between their time drinking the water or working with fluoropolymers and their half-lives of PFOA.

The monkey model was extrapolated to the human based on the hypothesis that the long-half lives in humans for PFOA and PFOS, like the monkey, can be attributed to saturable renal resorption in the proximal tubule of the kidney. We used the human model to successfully describe the available PFOA data from two communities exposed to PFOA in their tap water and also the general population. In addition, we used the model to predict the serum half-lives of 25 residents of the Little Hocking community.

Obviously, just because models that incorporate saturable renal resorption are able to describe the existing body of kinetic information does not mean that this process controls the serum half-lives of these chemicals in the animal or that the same process is responsible in the human. However, existing data for renal transporters in the rat suggest a tubular resorptive role for at least one transporter (Oatp1a1), and we were able to achieve good agreement between the observed data experimental data and the model predictions for both monkeys and humans. In addition, we have developed a PBPK model for these compounds in the rat using the renal resorption description, and this model successfully describes kinetics in the rat (manuscript in preparation). This is evidence that the saturable resorption process is responsible for the kinetics of these chemicals across species. If this process is confirmed to be the process responsible for the long half life in humans and the transporter proteins are identified, information on the activity of the transporters can be used to identify susceptible individuals due to polymorphic variation.

A recent study by Yang et al. showed that two human renal organic anion transporters on the apical membrane side of proximal tubule cells are capable of transporting PFCs and thus may play a key role in tubular resorption of these compounds (Yang et al., 2010). This further suggests that renal resorption is at least partly responsible for the long plasma half-life of these chemicals. Two other human OATs, OAT1 and OAT3, which are located on the basolateral membrane of proximal tubule cells, can also transport PFCs (tubular secretion) (Weaver et al., 2009); OAT1 and OAT3 thus excrete the chemicals from the blood to the urine side, while OAT4

and URAT1 bring the chemicals back into the cell. In the rat, OAT1 and OAT3 have been shown to secrete PFCs; the higher expression of OATp1a1 (apical side) in the male rat is related to the longer half-life observed in the male rat. The human orthologs of OAT1 and OAT3 have similar affinities for PFCs and due to the high expression levels of these transporters in the human, tubular secretion is expected to be similar to that in the rat (Nakagawa et al., 2008) (Hilgendorf et al., 2007; Nakagawa et al., 2009). However, because PFCs do not accumulate in the kidney, Yang, et al. hypothesize that there may be an efflux transporter localized to the basolateral membrane that could aid in transporting PFCs back into systemic circulation. (Yang et al., 2010) If identified, this efflux transporter(s) would, in addition to the apical transporters responsible for tubular resorption, probably contribute to extending the half-life of PFCs in humans.

4.1. Monkey PBPK models

In the oral dosing study with PFOA, it was observed that there is a rapid rise to steady state plasma concentrations of PFOA with a much slower terminal half-life (Butenhoff et al., 2002, 2004a). In the studies with PFOS, a similar trend is observed, and the terminal half-life is even slower (Seacat et al., 2002). In addition, the differences observed in elimination with increasing dose are indicative that a capacity-limited, saturable process must be involved in the pharmacokinetic behavior of these compounds (Andersen et al., 2006; Tan et al., 2008). It is hypothesized that saturable renal resorption is the process responsible for the kinetics and long half-lives of these chemicals; thus, this description was incorporated into both the compartmental model of Andersen and Tan and the current PBPK model for PFOA and PFOS. This process was described with a transporter maximum (T_m) and transporter affinity constant (K_t) where chemical was reabsorbed from the filtrate compartment back to the kidney into the plasma. The longer half-life of PFOS compared to PFOA was accounted for by using a higher capacity and higher affinity for PFOS in the model. The biologically motivated model required a time-dependent feature to better characterize the kinetics of these chemicals, which was a changing free fraction of chemical in the plasma. This feature was not necessary for description of kinetics in the PBPK model; a constant free fraction (F_{free}) was sufficient. Because both PFOA and PFOS are highly bound to plasma proteins (>97%), a free fraction of <0.03 (0.02 for PFOA and 0.025 for PFOS) was estimated to fit the plasma time course data for the PBPK model. The monkey models successfully describe the available PK data, and further improvement to the monkey PBPK model will rely on additional data. Information on plasma binding proteins in the monkey (dissociation constants, number of binding sites for PFOA and PFOS) and elucidation and information (any sex differences in expression, transporter capacities and affinity constants) on renal transporters responsible for tubular resorption in the monkey would aid in refinement of the model.

The goal of developing a PBPK model for the monkey was to extrapolate to the human in order to enhance the scientific basis for human health risk assessment of PFOA and PFOS. Due to the physiological structure of the monkey model, it was possible and easy to extrapolate the model to the human. Epidemiological data suggest that the half-life of PFOA and PFOS is on the order of years in the human in contrast to months in the monkey. If renal resorption is the main process responsible for the long half-life of these compounds in both species, the slower clearance in the human can now be easily described in the model by a higher transporter capacity. In addition to extrapolation to the human, the PBPK model for the monkey could be extended to describe the kinetic data available for other perfluoroalkyl acids.

4.2. Human PBPK models

Available human data for PFOA and PFOS suggest that humans are slow eliminators of these chemicals. It is hypothesized that these long half-lives are due to saturable renal resorption as they are in the monkey and rodent. The human PBPK model thus incorporated saturable renal resorption in the kidney and exhibits good agreement with available data for PFOA in Little Hocking and Arnsberg residents and also in the general population. Human data on PFOS is even sparser than that for PFOA, but using the half-life estimate from retired workers and the data available on serum PFOS concentrations from Little Hocking, the model shows reasonable agreement with the data. Plasma protein binding is another possible cause of the long plasma half-lives in humans. PFOA and PFOS have been found to be >90% bound to plasma proteins in rats, monkeys, and humans; however, binding is not well-characterized (Han et al., 2003; Jones et al., 2003; Kerstner-Wood et al., 2003). For example, it may be difficult to distinguish between 90% and 99% bound, but if humans have a higher bound fraction, this would mean a 10-fold difference in the free fraction and therefore a slower clearance and longer half-life would be possible. Further refinement will depend on future studies of both renal transporters and plasma protein binding.

Even though the model exhibits good predictive ability, we had only means, medians, and standard deviations of human serum measurements at our disposal. We were not able to obtain the data on the individuals in the populations in the studies, so we were not able to incorporate factors that may be important to determining variability in serum concentration, plasma half-life, or susceptibility to toxic effects. These factors would include, but may not be limited to: gender, age, how long each person had been drinking the water, how much tap water each person drinks daily, the PFOA concentration measured in different wells or water supplies, and any preexisting medical conditions, such as compromised renal function. The more data we can obtain on individuals in exposed populations, the better we can characterize population variability in a risk assessment context. Several years ago, residents from Little Hocking and the surrounding communities of the fluoropolymer production plant filed a class-action lawsuit alleging adverse health effects due to consumption of water containing PFOA. As part of the settlement, the C8 Health Project, which is an independent panel of researchers investigating the potential link between PFOA exposure and adverse health effects, was created. We believe that using this information when it becomes available to us would greatly improve the value of the model for describing the variability in the human pharmacokinetics of PFOA.

When running simulations, we took into consideration both serum elimination half-life estimates available in the literature for PFOA. From the model predictions, both half-lives appear to be equally likely. Both populations used to estimate the half-lives were small (26 for workers and 200 for environmental exposure); however, the 26 workers exhibited half-lives from 1.5 to 9 years (the environmental study did not give the individuals' half-lives). From the half-lives that the model predicted from the 25 residents in Little Hocking, the half-lives range from 0.9 to 8 years. Even in small populations a wide spectrum is observed, indicating that there is probably a lot of variability in how individuals retain, handle, and eliminate PFOA. The average half-life from the model predictions was 2.7 years, which is between that measured from the workers (3.8 years) and the environmentally exposed people (2.3 years).

Future refinement of the human model will depend on additional human studies and information on activity and variation in renal transporters. Use of the human PBPK model along with individual data and information on exposure routes and sources of PFAAs will enable a better understanding of pharmacokinetics of

PFAAs in humans, link possible exposure routes and external doses to blood concentrations, aid in understanding interindividual variability, and ultimately aid in human health risk assessment for PFAAs.

Conflict of interest statement

The authors declare that there are no conflicts of interest.

Acknowledgments

We would like to thank Dr. John Butenhoff at 3M Company and Mr. Gerald Kennedy, Jr. at DuPont Company for helpful discussion and comments. Financial support was supported by an EPA STAR Grant (No. R833450) and the 3M and DuPont Companies.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.yrtph.2010.12.004.

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